

01_MME_MJJ_2025

April 28, 2025

#####



#

AGHYMET RCC-WAS

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0.1 Library

Set forecast working directory

Set Climatological Years

0.2 Download Observation and Process

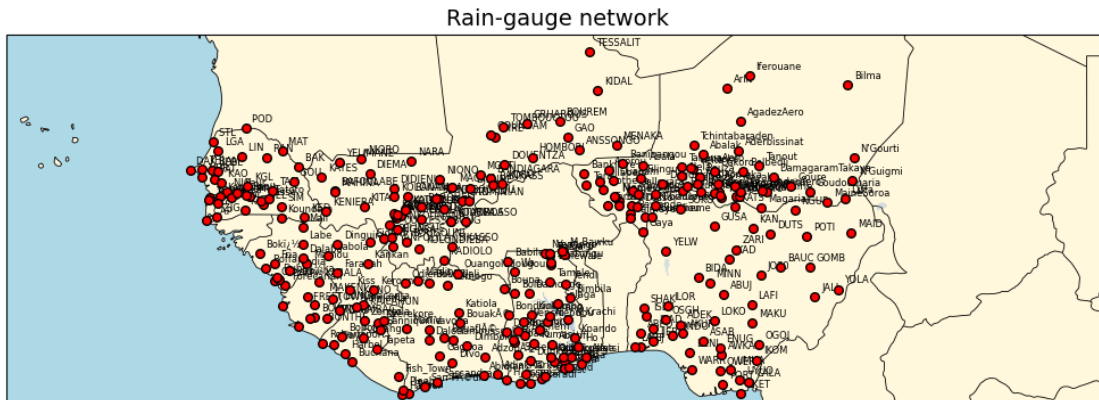
Load Downloader

Data Download Area



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Process Obs in combining with Seasonal cumulative Ground-Observation



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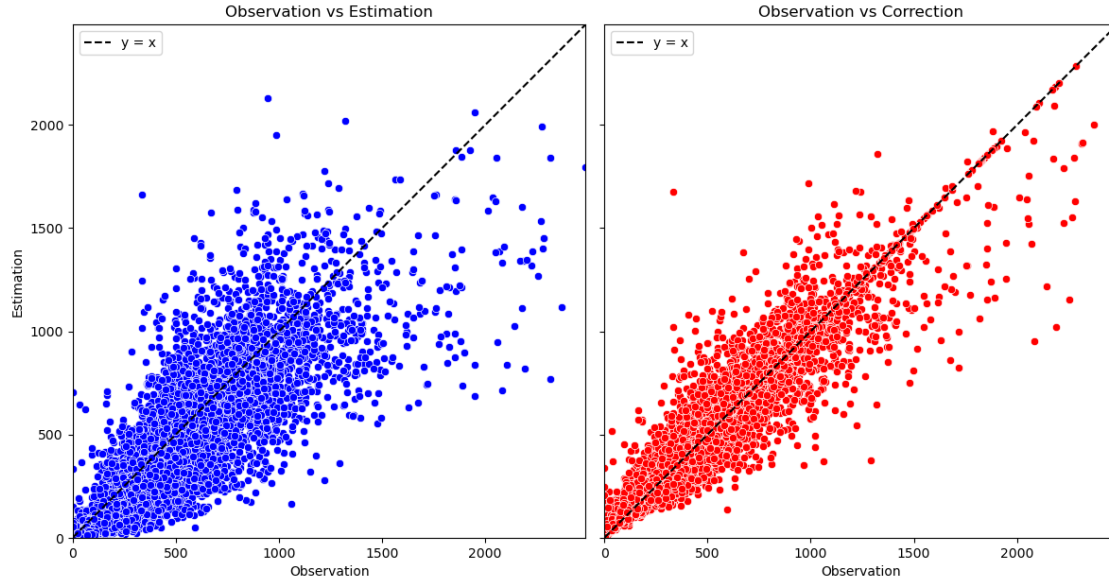
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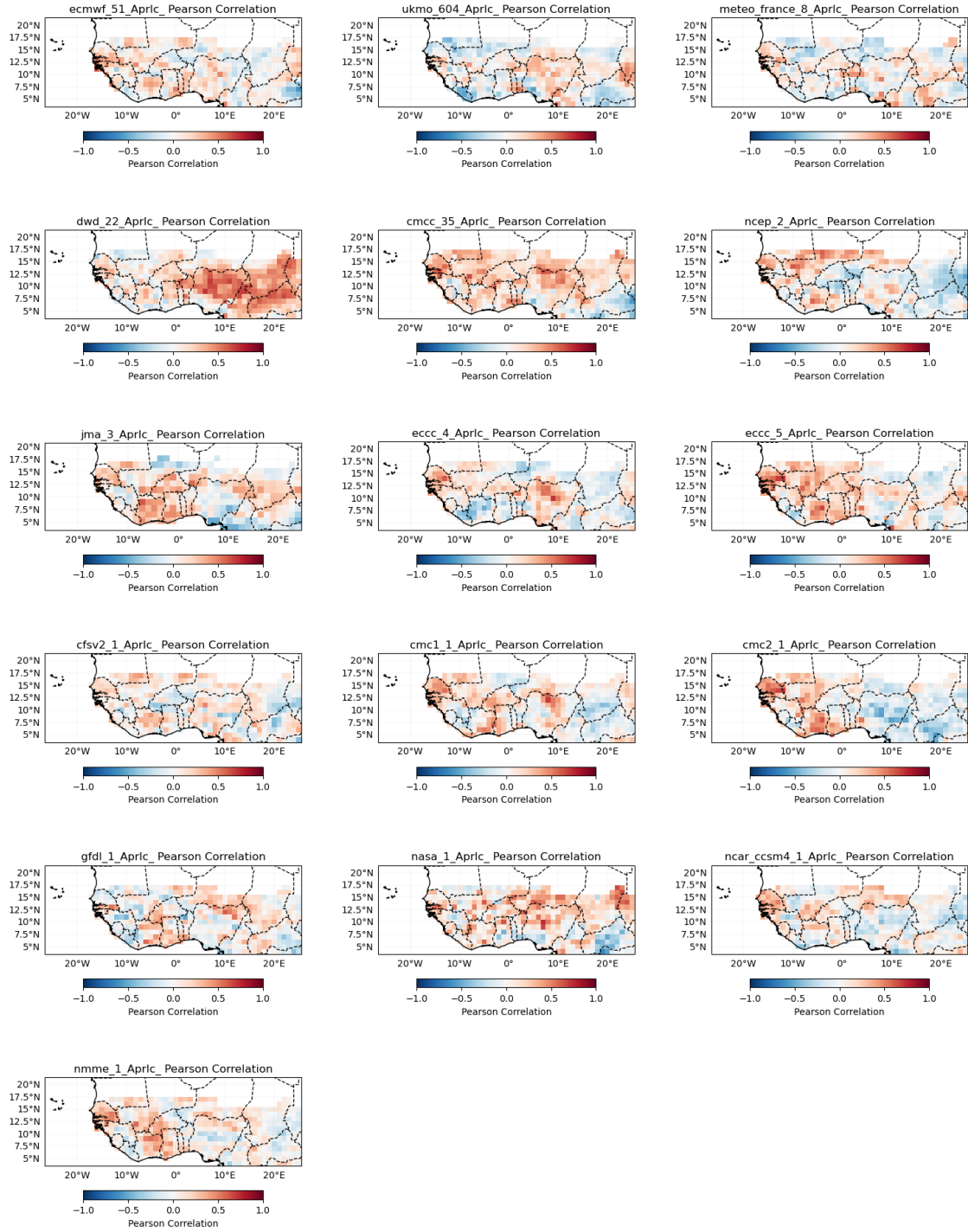
1 1st - Approach: Choice of Best Dynamical models and weighted Average

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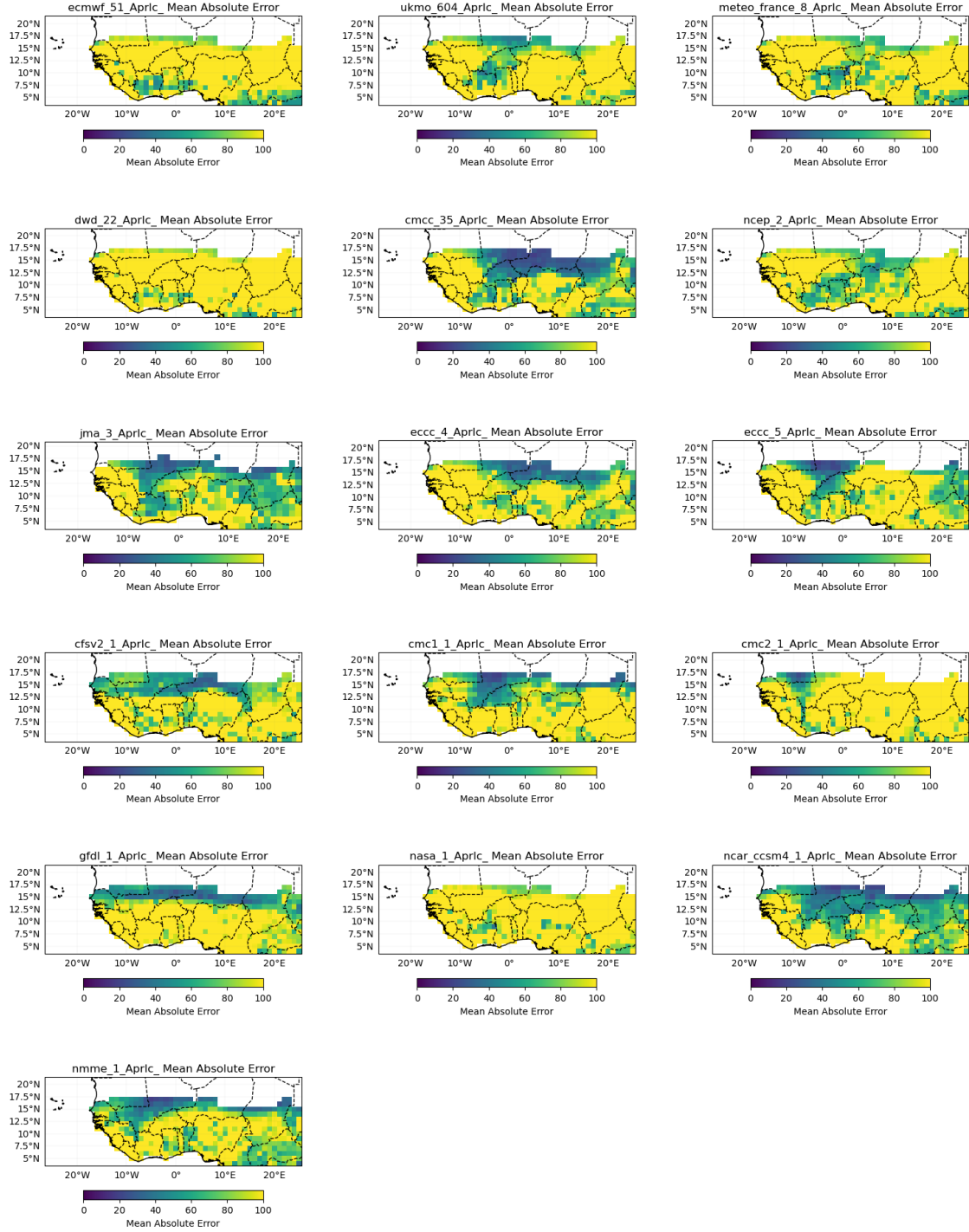
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Validation of GCM

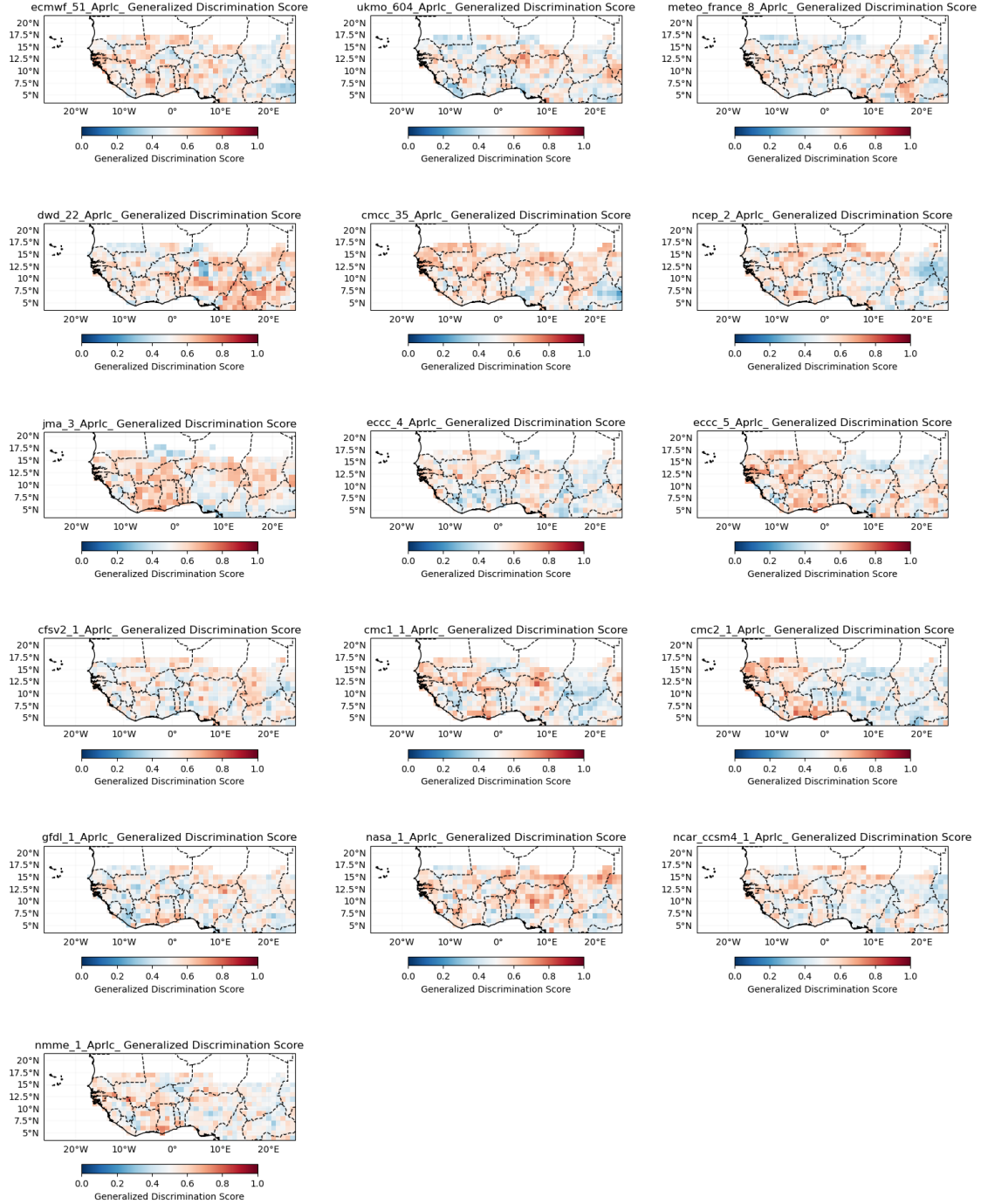
Pearson Correlation



Mean Absolute Error



Generalized Discrimination Score



1.1 Choice of best GCM models

`['CMCC_35.PRCP', 'NASA_1.PRCP', 'DWD_22.PRCP', 'ECCC_5.PRCP']`

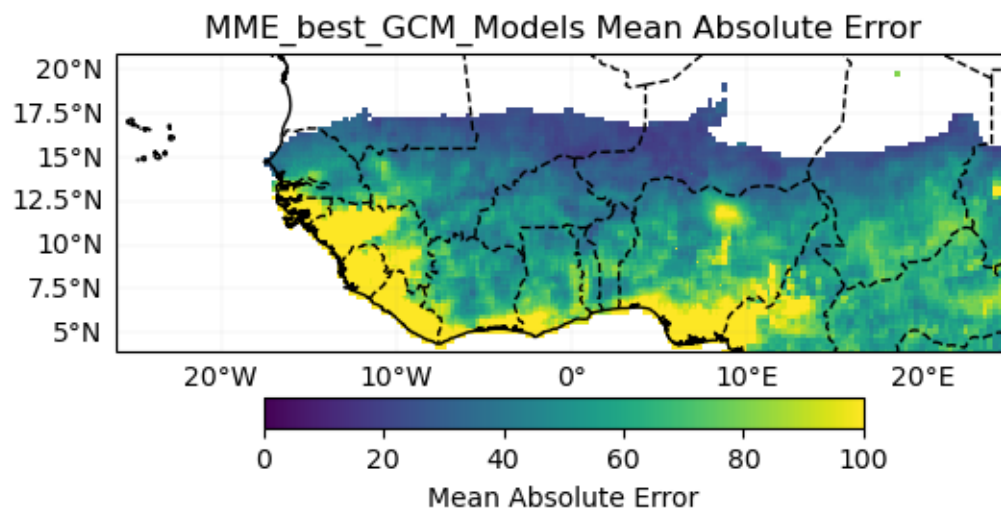
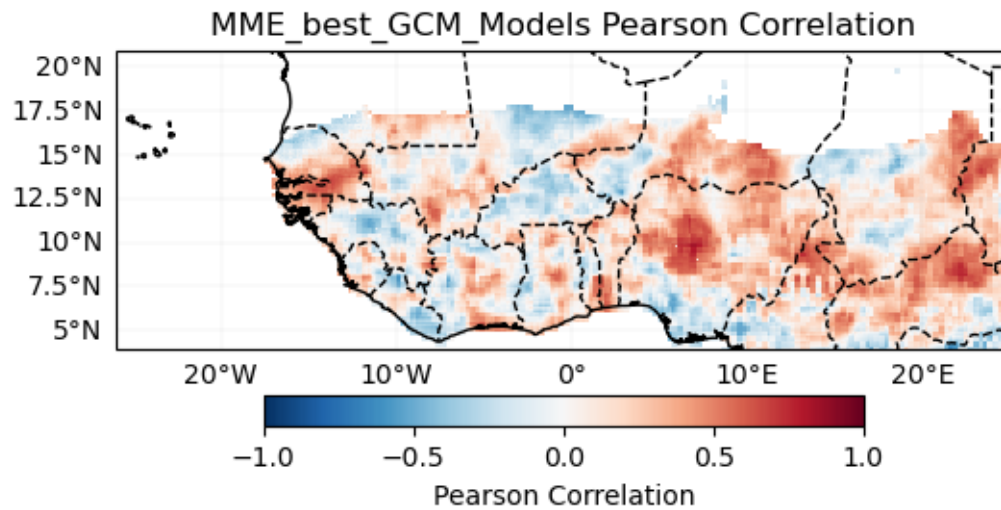
1.2 MME for best GCM models

Cross-validation

Cross-validation ongoing

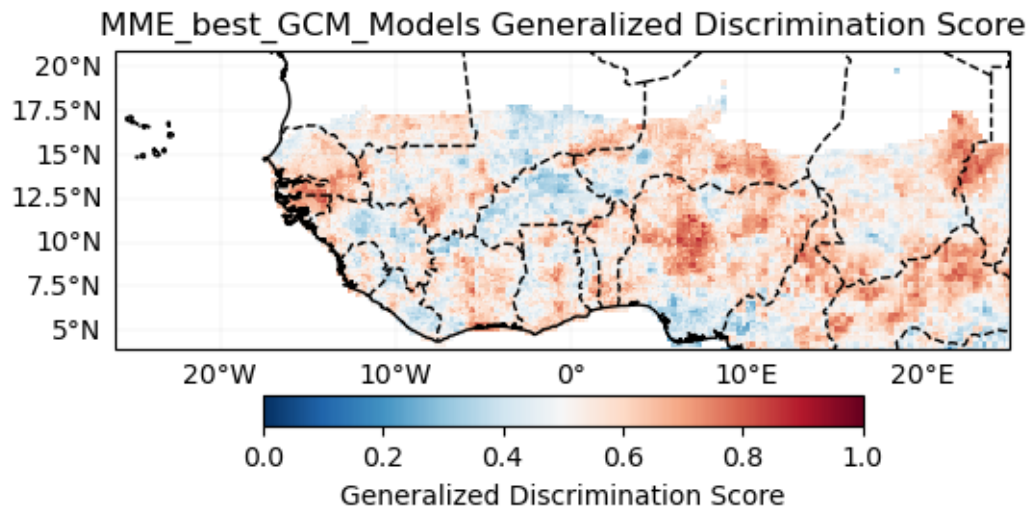
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Verification



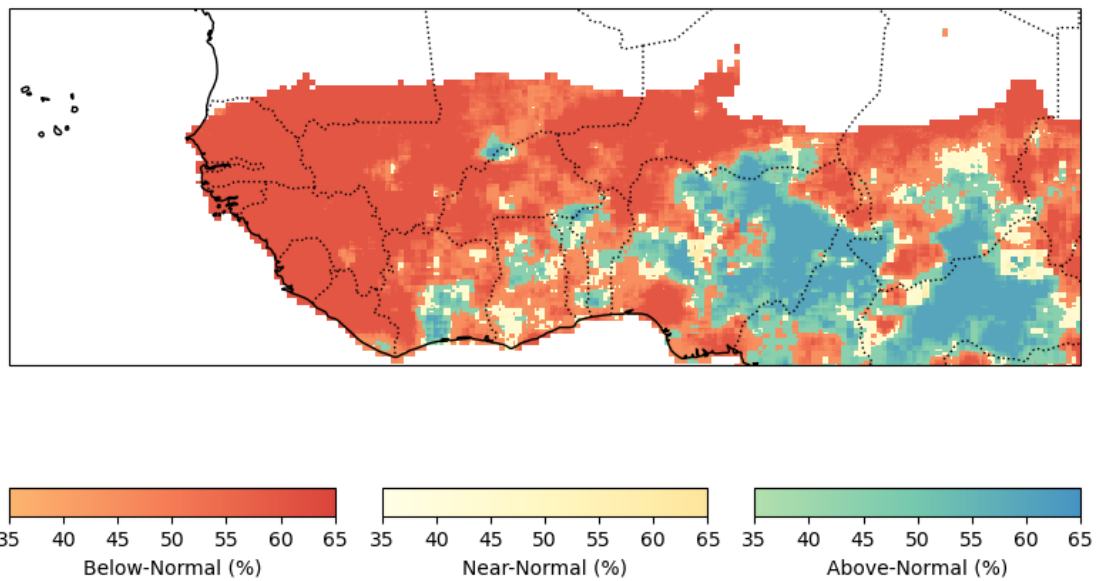
Probabilistic

Generalized Discrimination Score



Forecast

Probabilistic Forecast - MME_best_GCM_Models MayJunJul-2025



8549

2 2nd - Approach: Statistico-Dynamical

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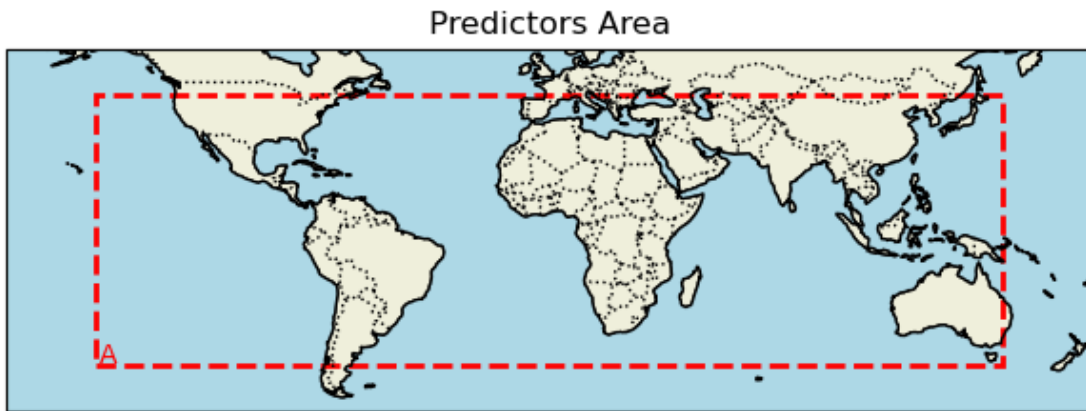
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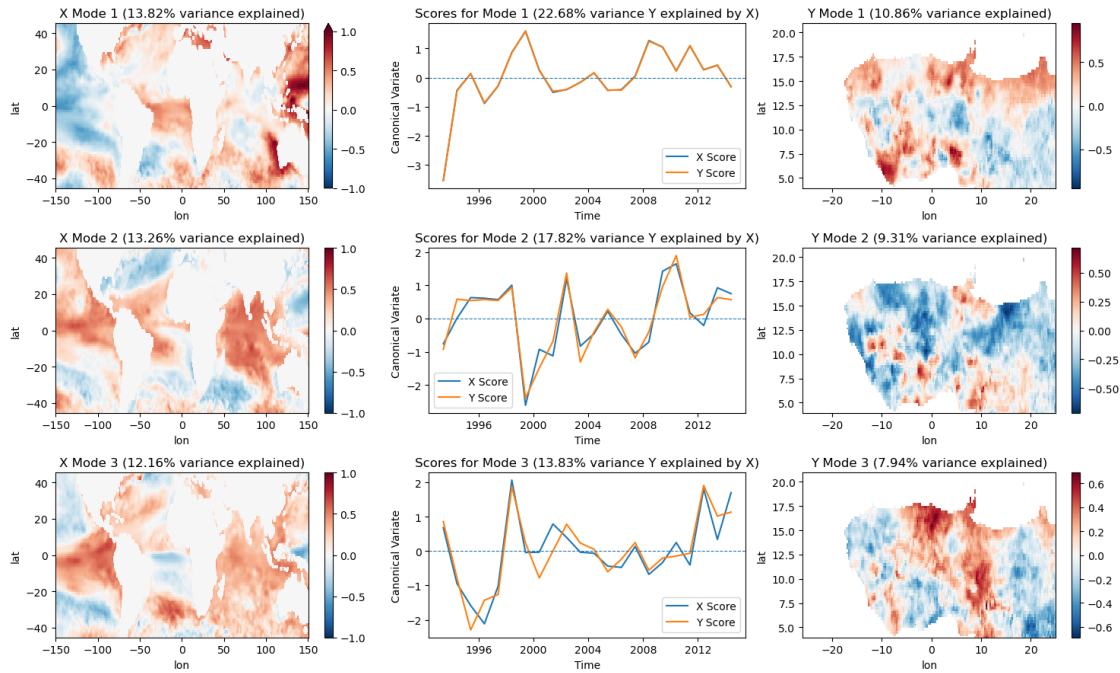
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2.0.1 Initialize the WAS_CCA class (default)

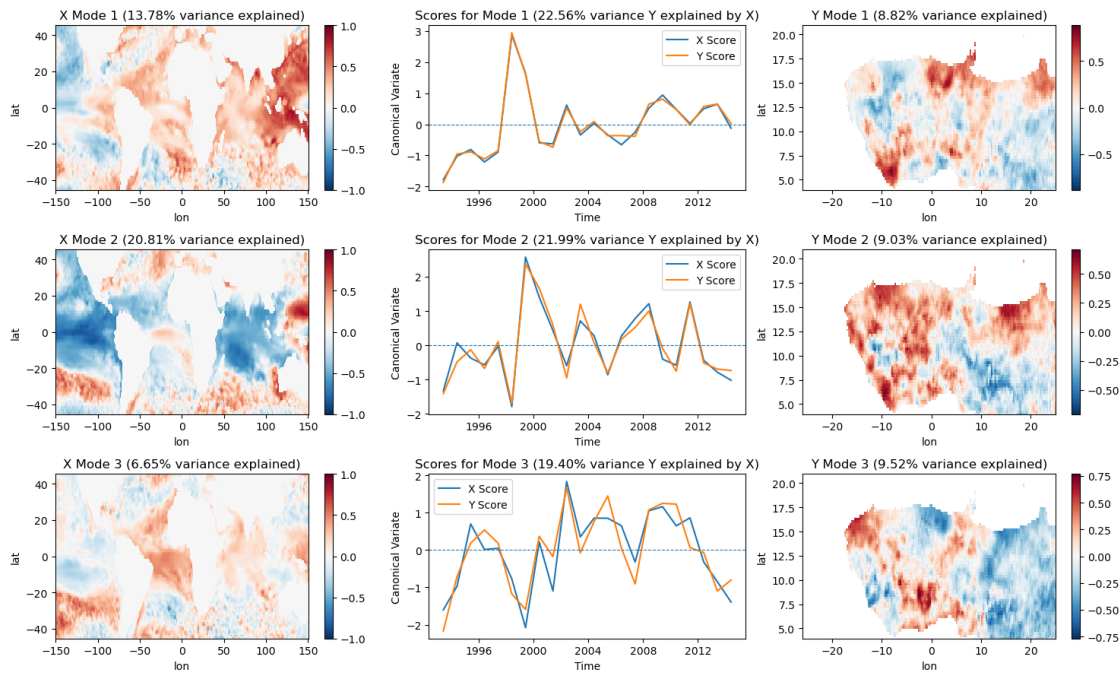


Modes of CCA for each model

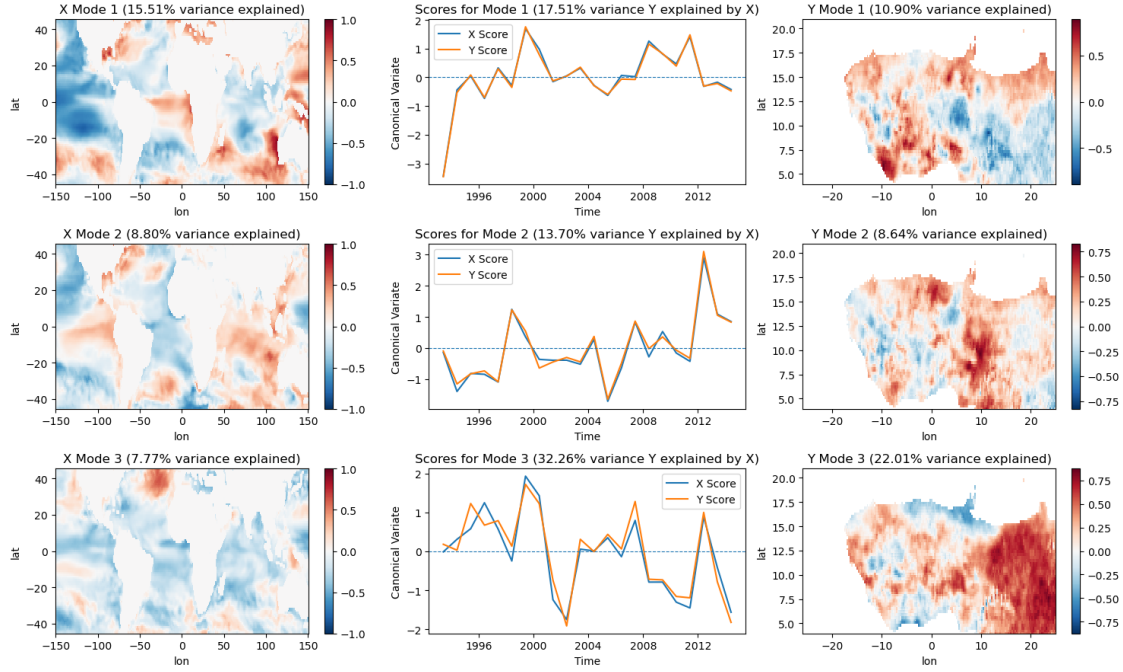
CCA Modes for model ECMWF_51.SST



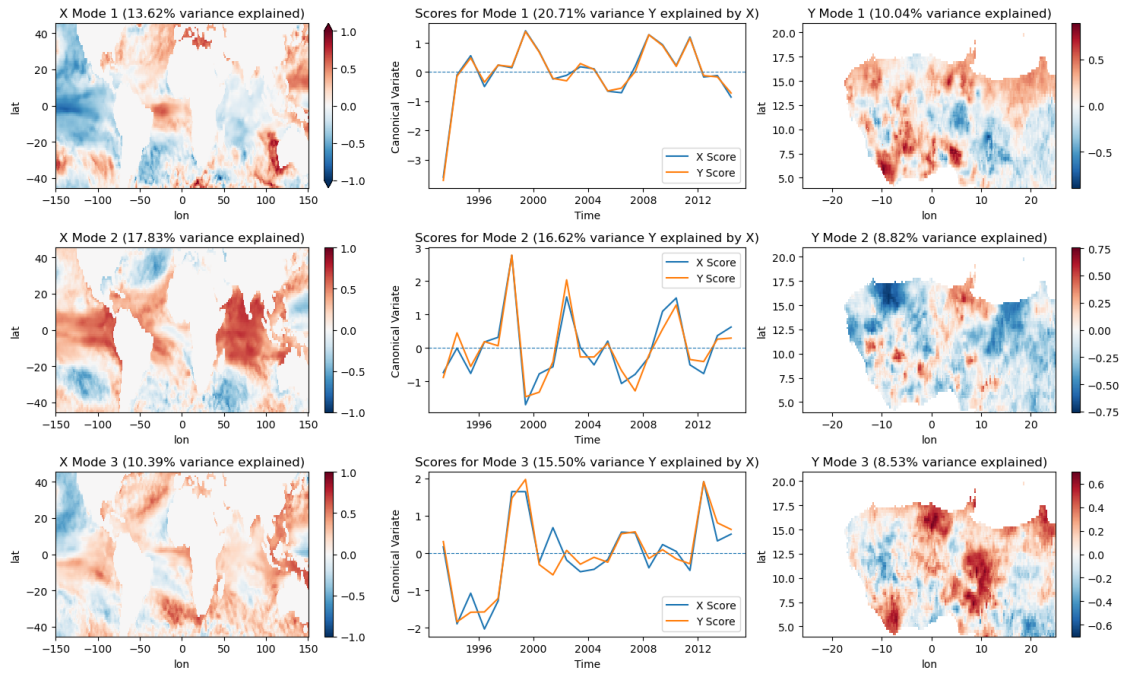
CCA Modes for model METEOFRACTE_8.SST



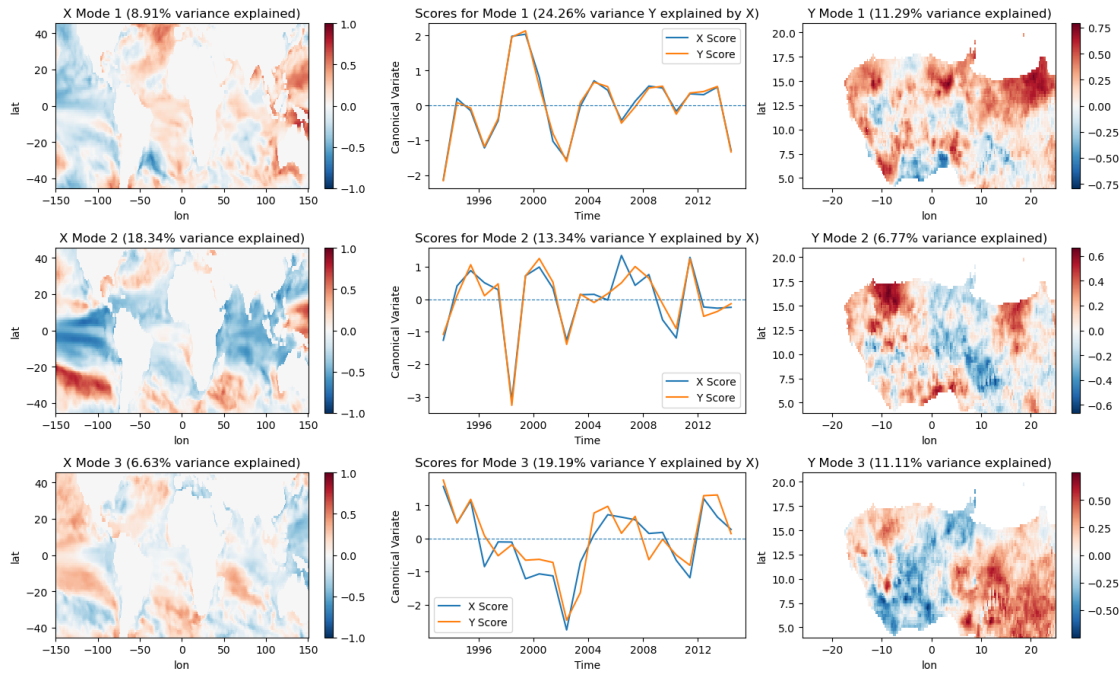
CCA Modes for model DWD_22.SST



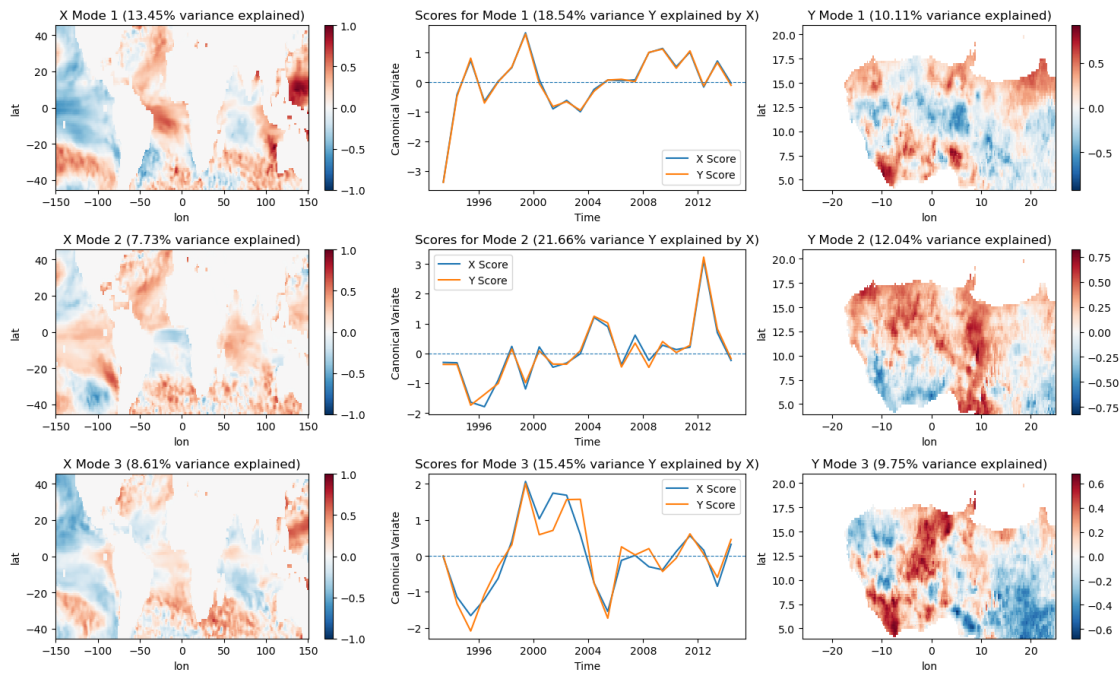
CCA Modes for model CMCC_35.SST



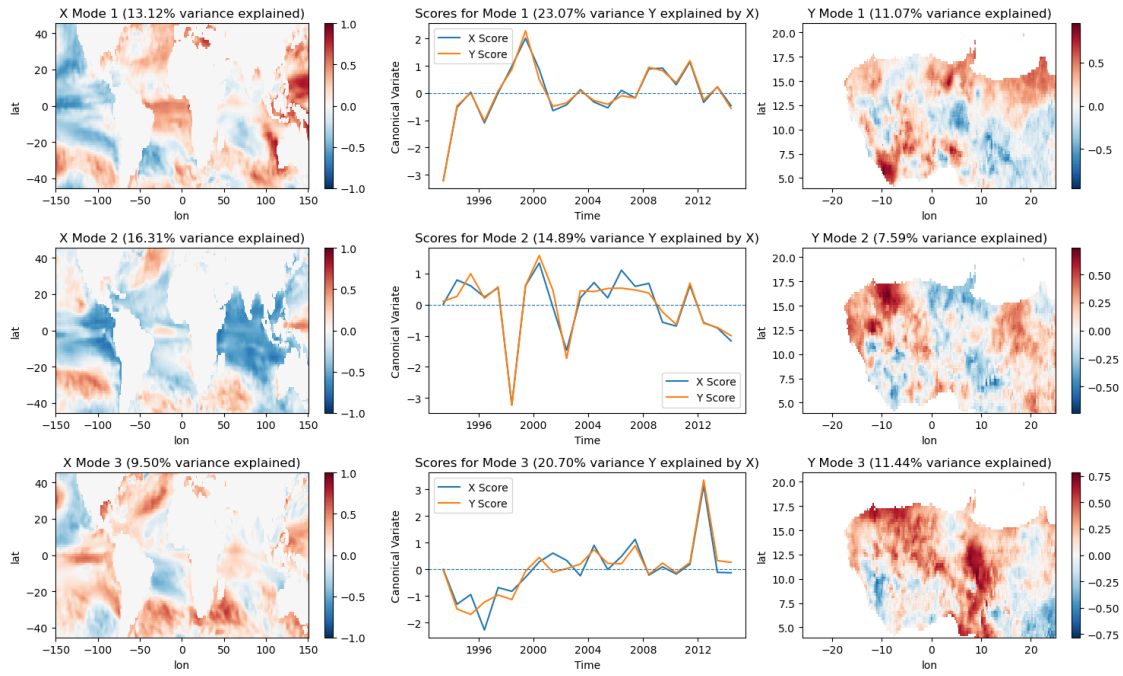
CCA Modes for model NCEP_2.SST



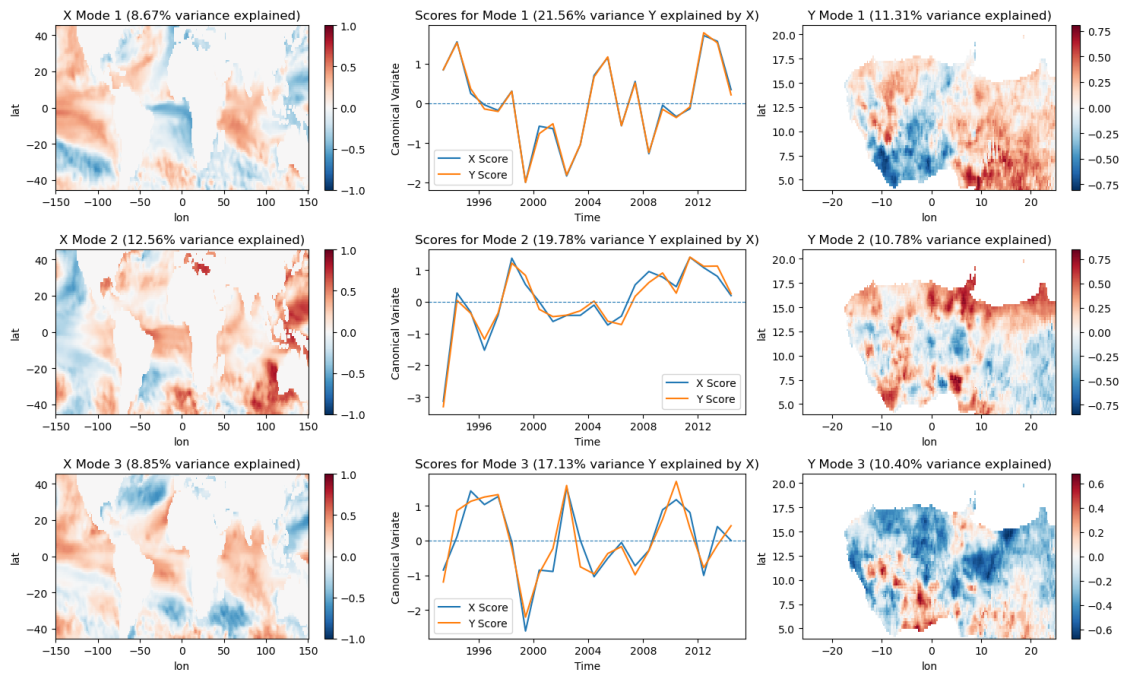
CCA Modes for model JMA_3.SST



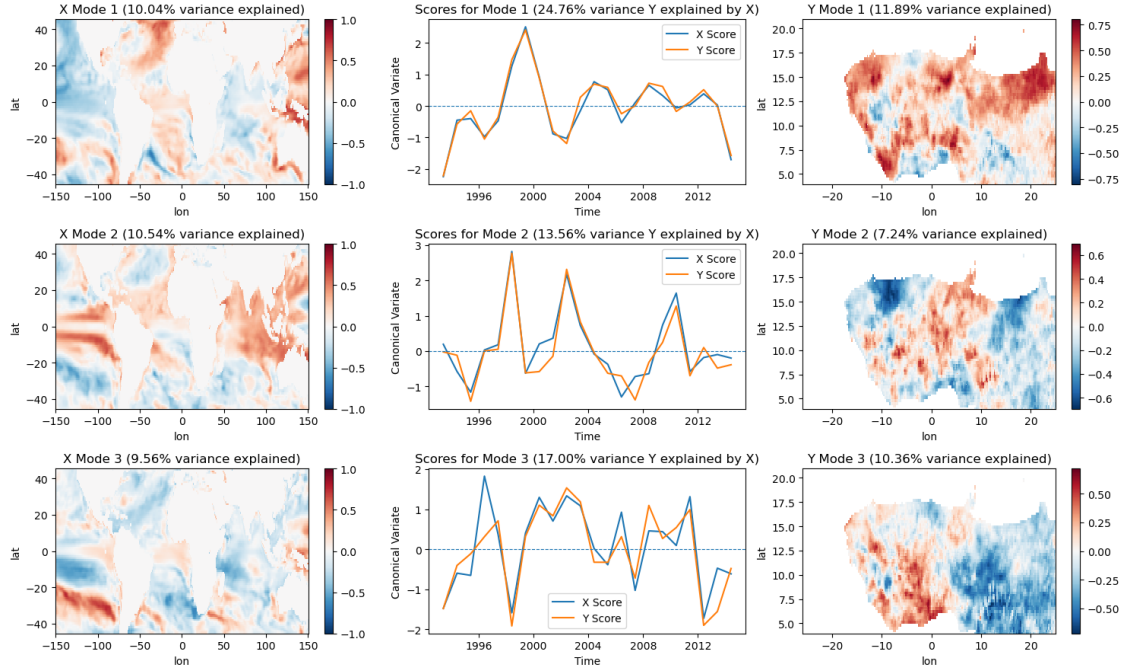
CCA Modes for model ECCC_4.SST



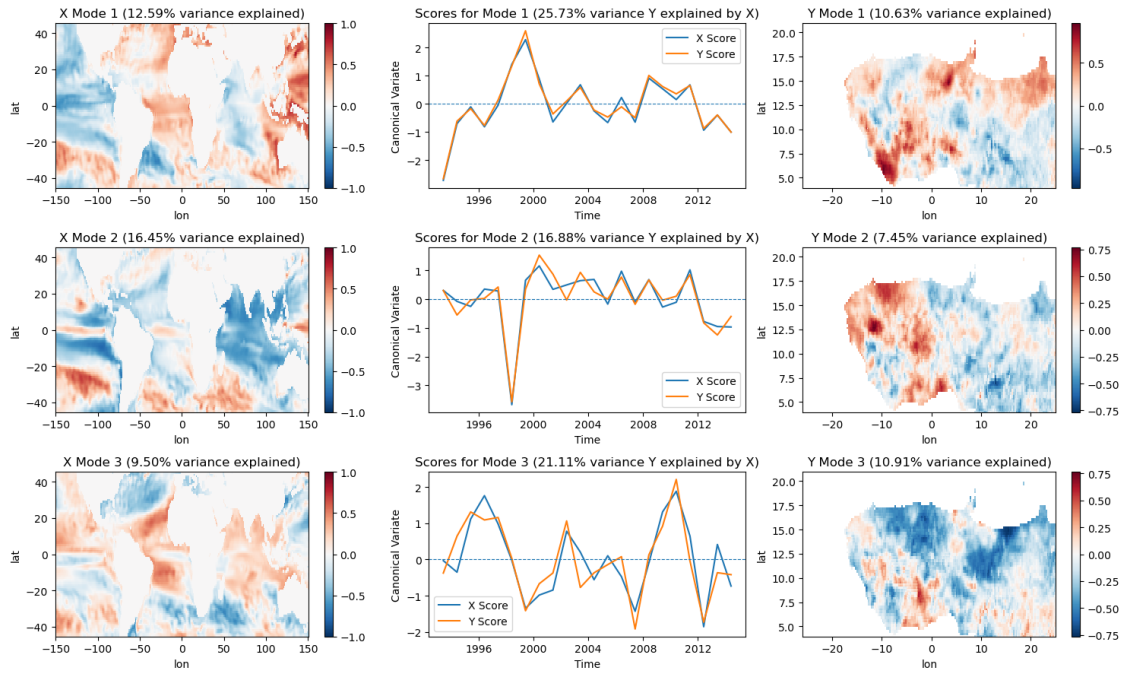
CCA Modes for model ECCC_5.SST



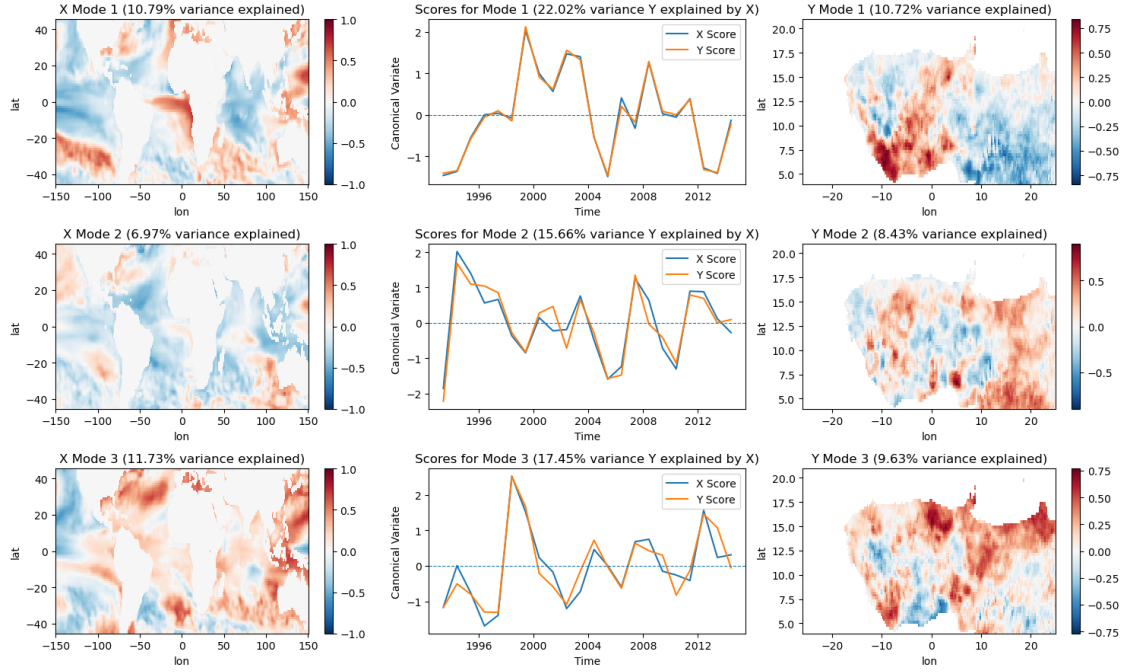
CCA Modes for model CFSV2_1.SST



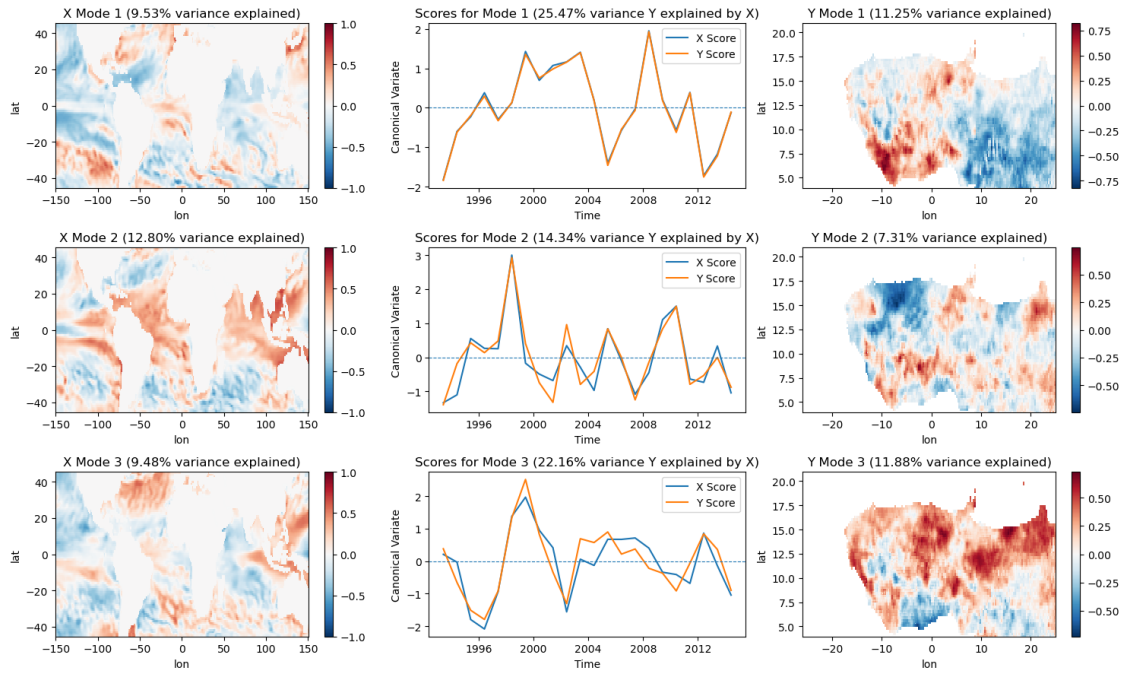
CCA Modes for model CMC1_1.SST



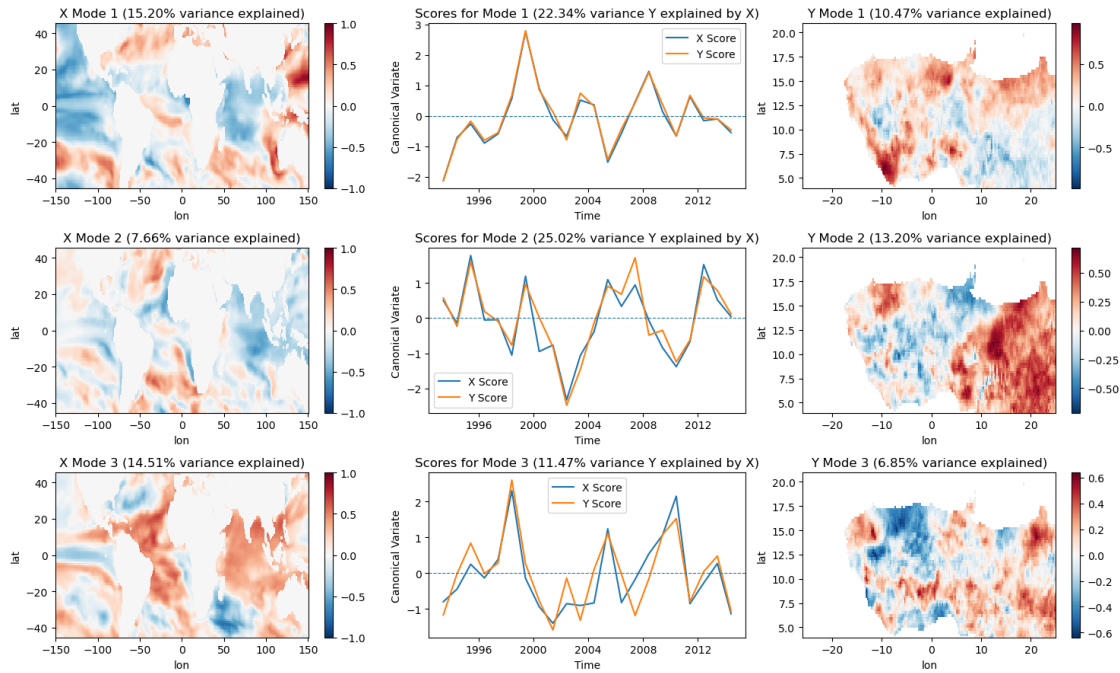
CCA Modes for model CMC2_1.SST



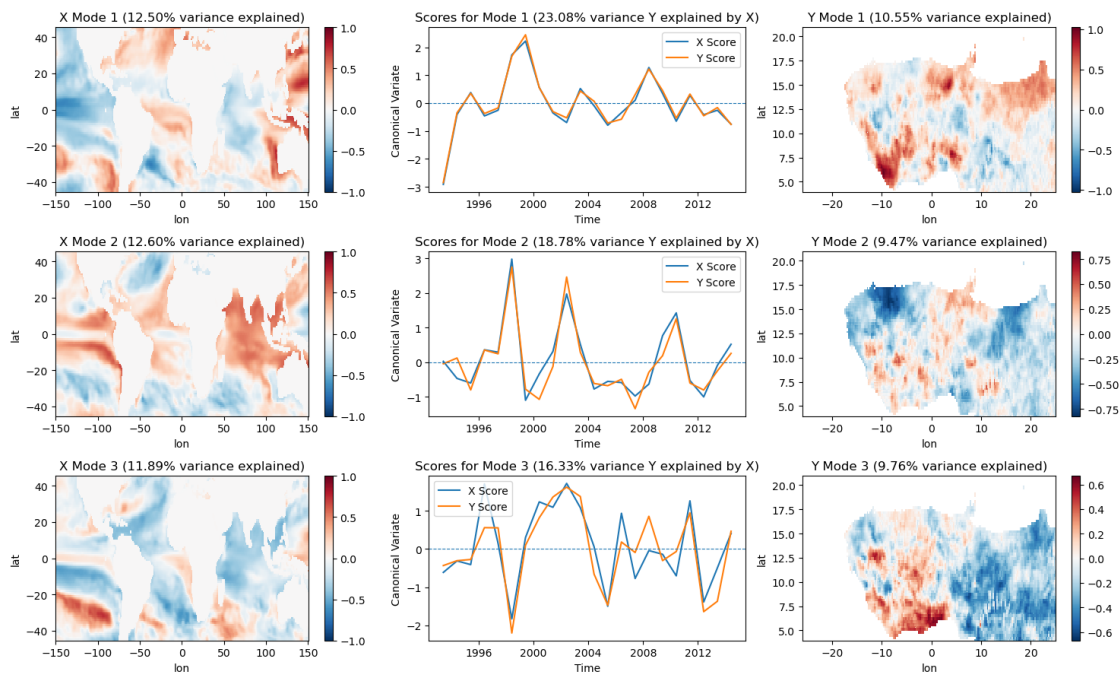
CCA Modes for model GFDL_1.SST



CCA Modes for model NCAR_CCSM4_1.SST



CCA Modes for model NMME_1.SST



Cross validation

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DWD_22		
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CMCC_35		
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NCEP_2		
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GFDL_1		
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NCAR_CCSM4_1

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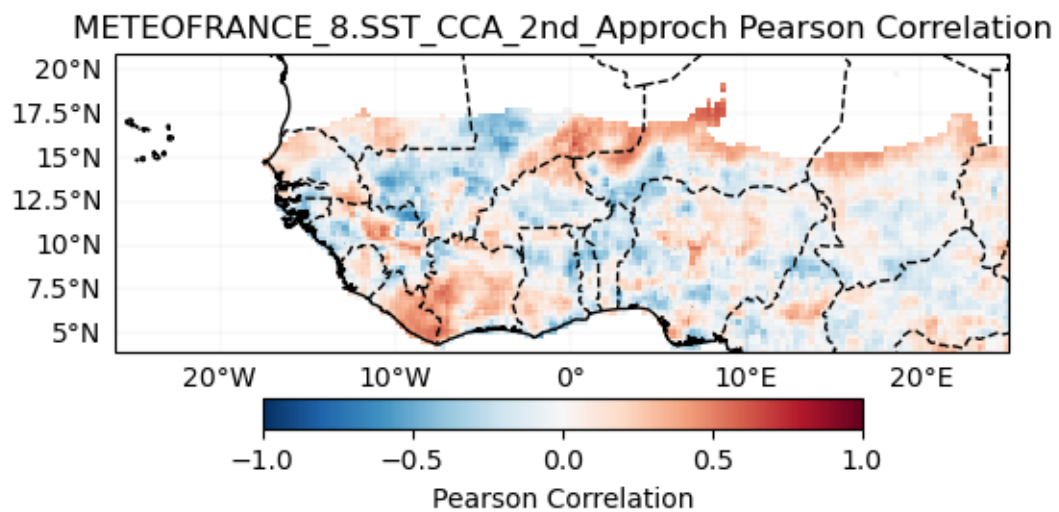
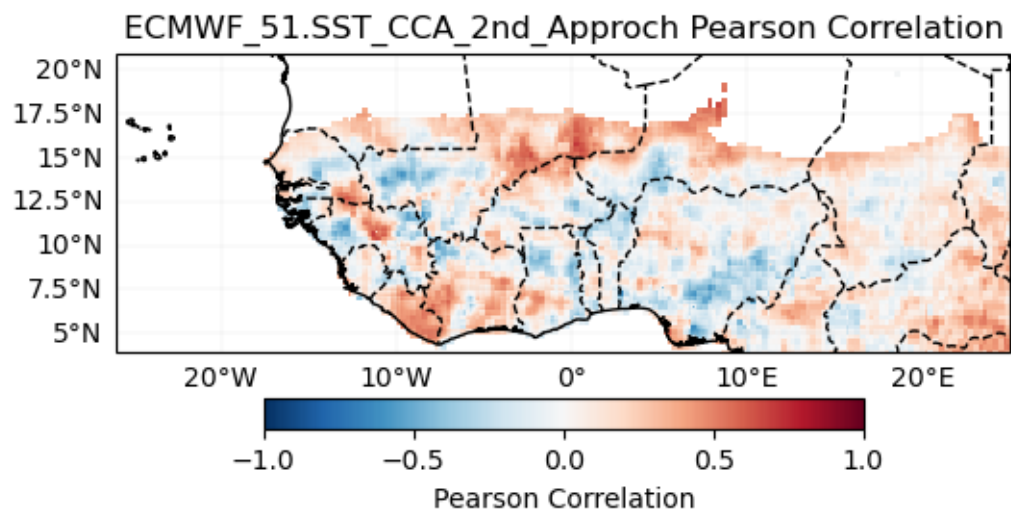
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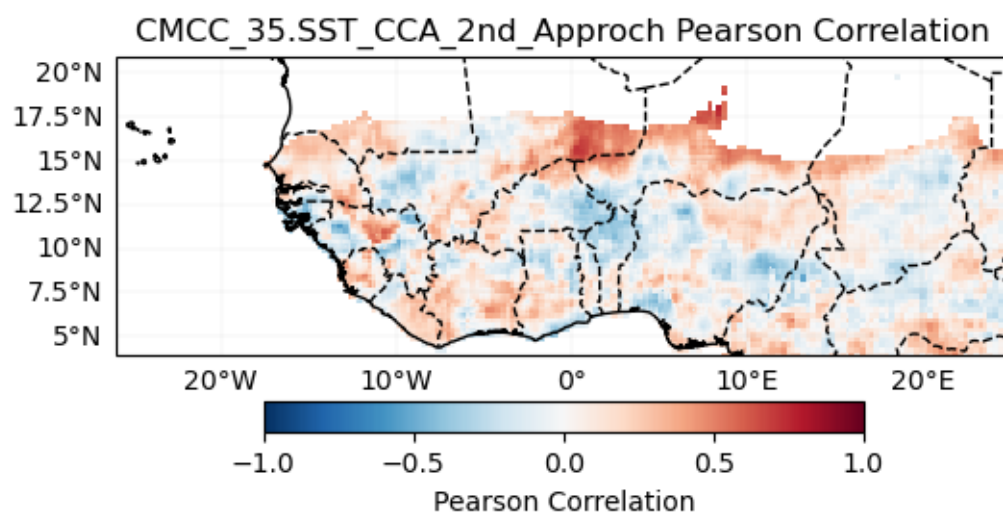
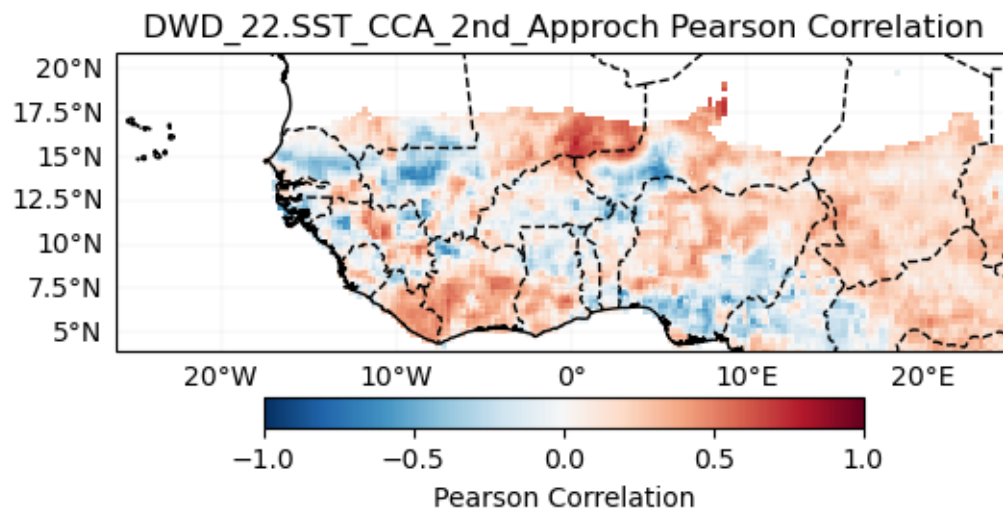
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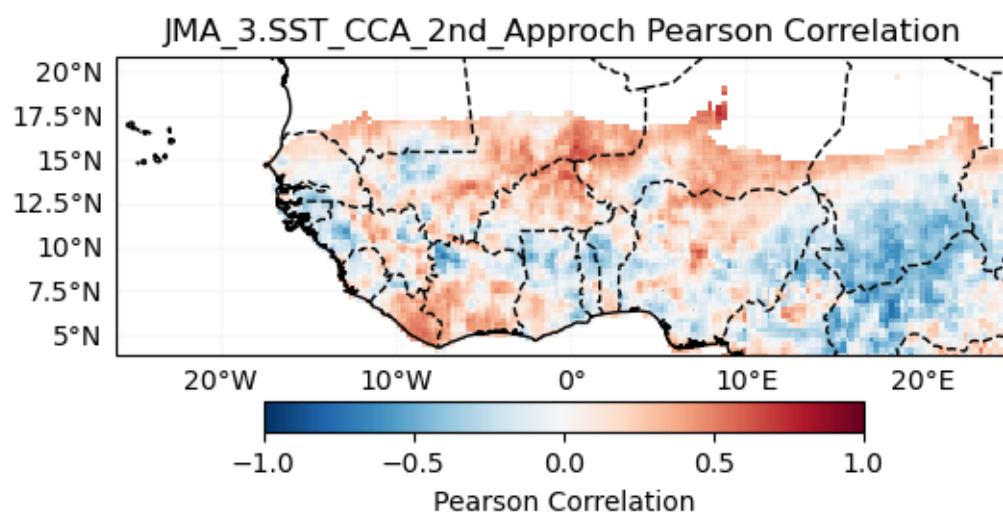
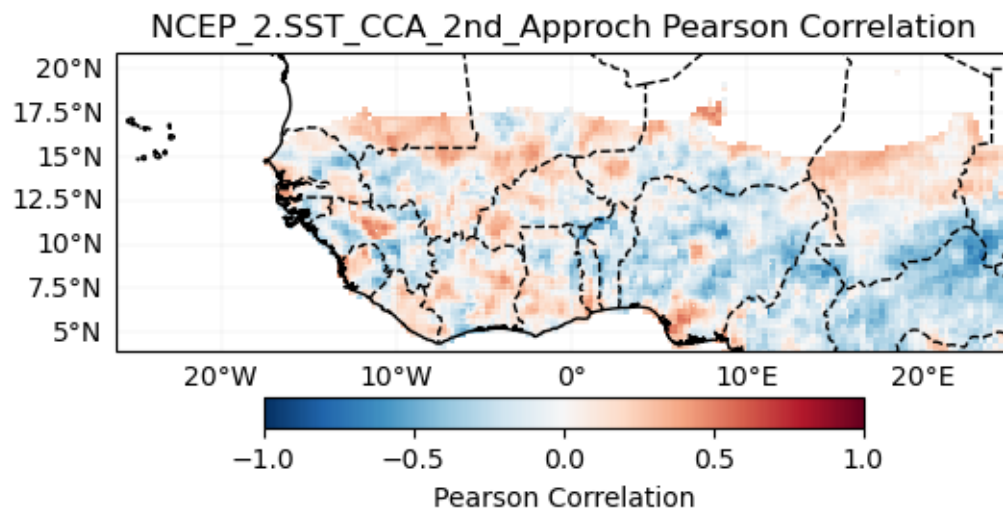
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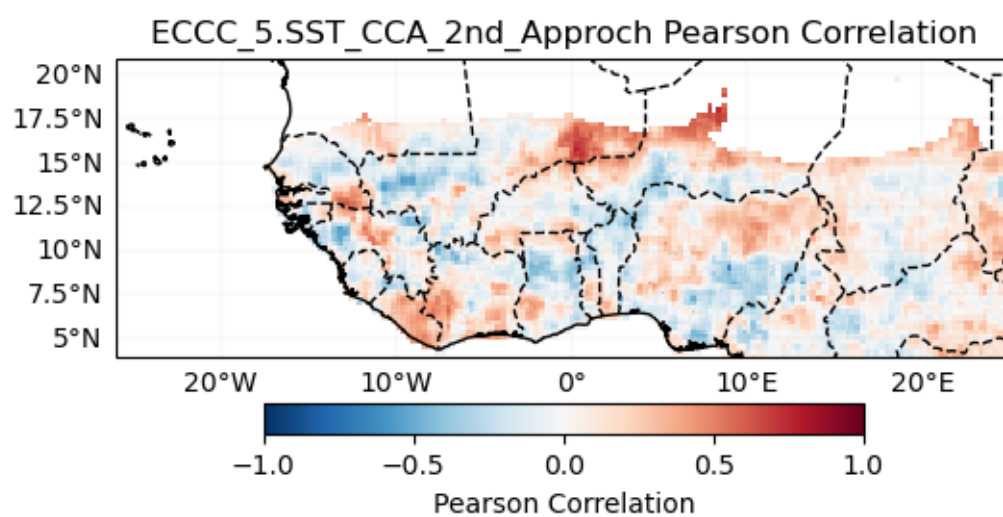
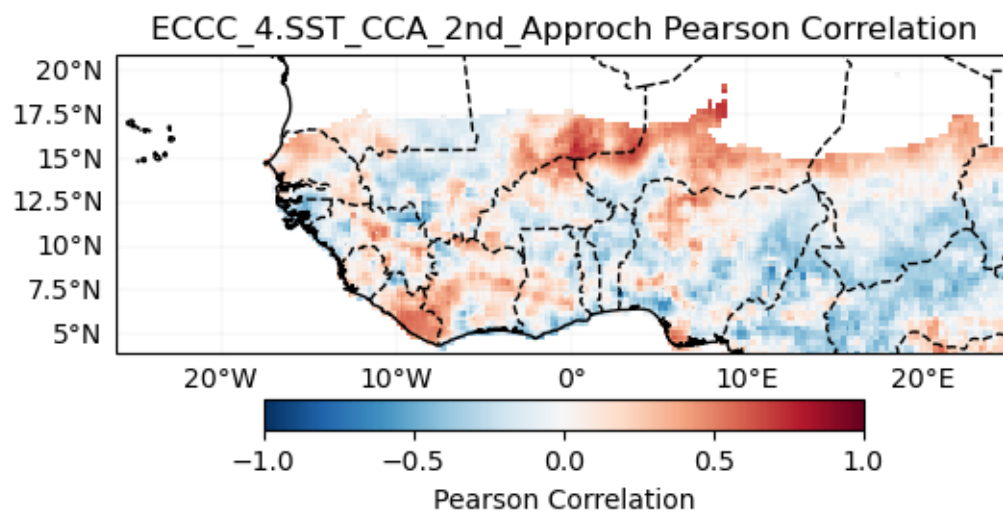
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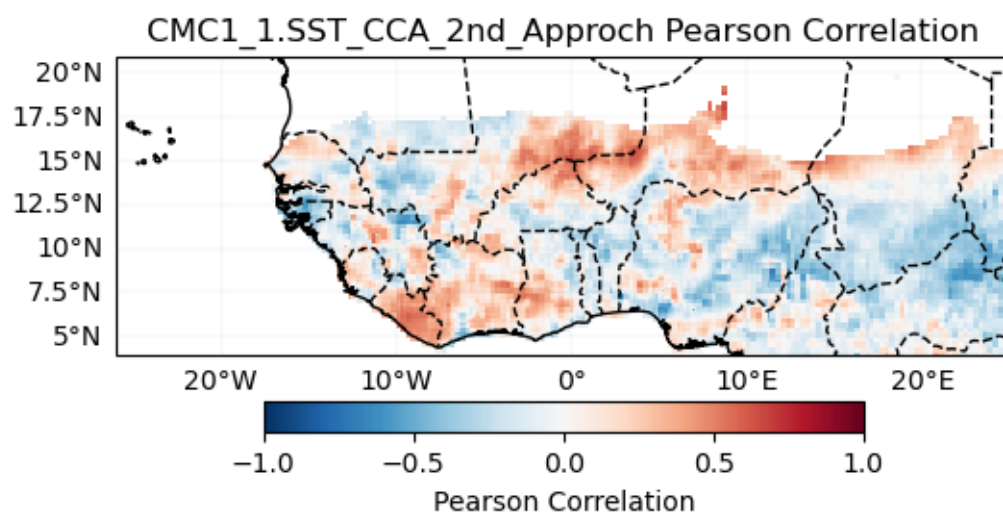
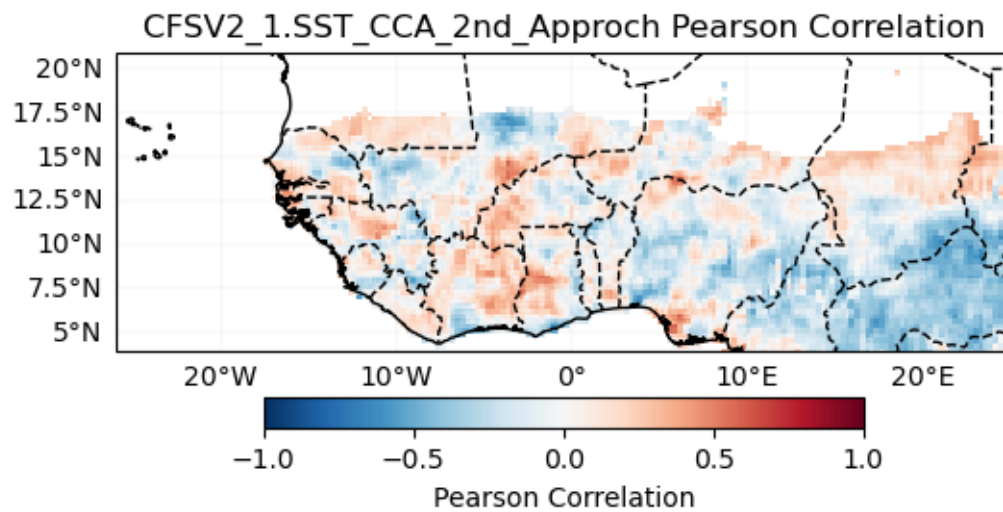
Pearson Correlation

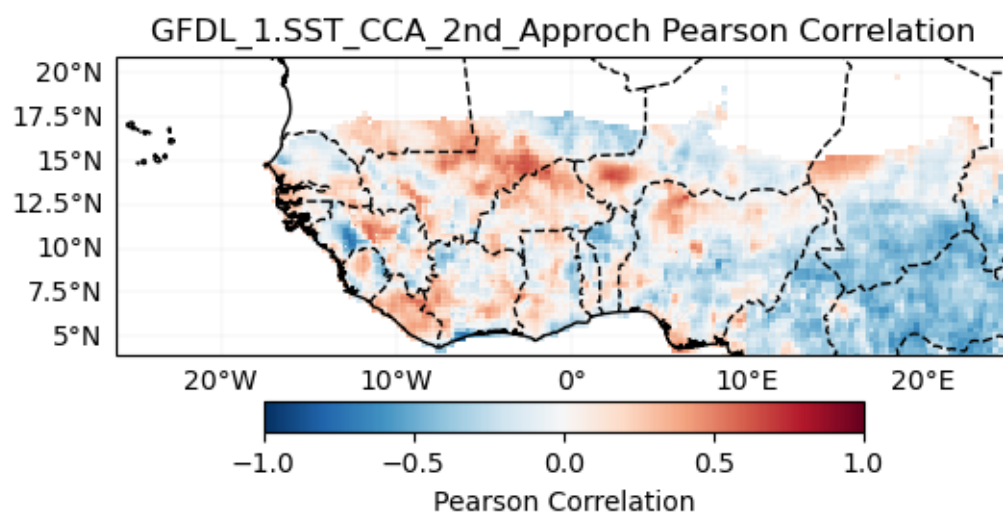
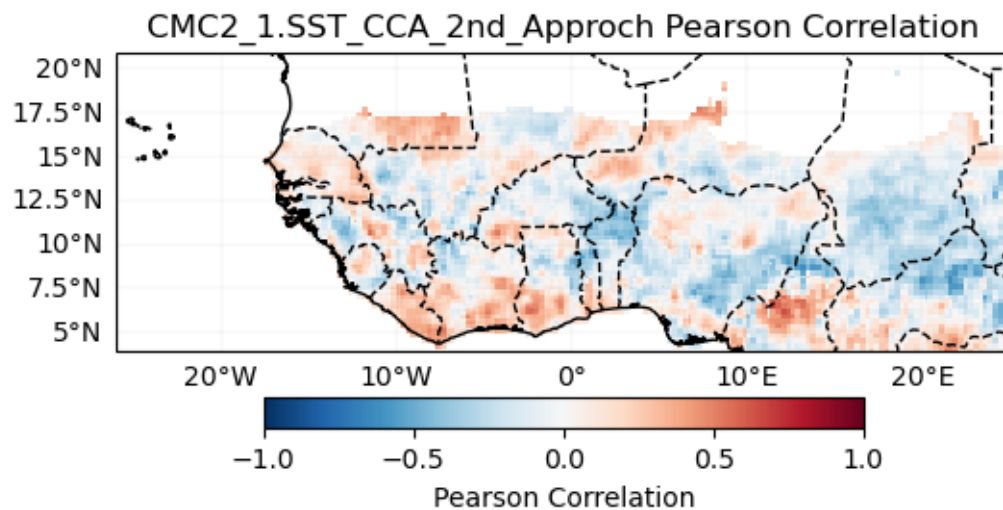


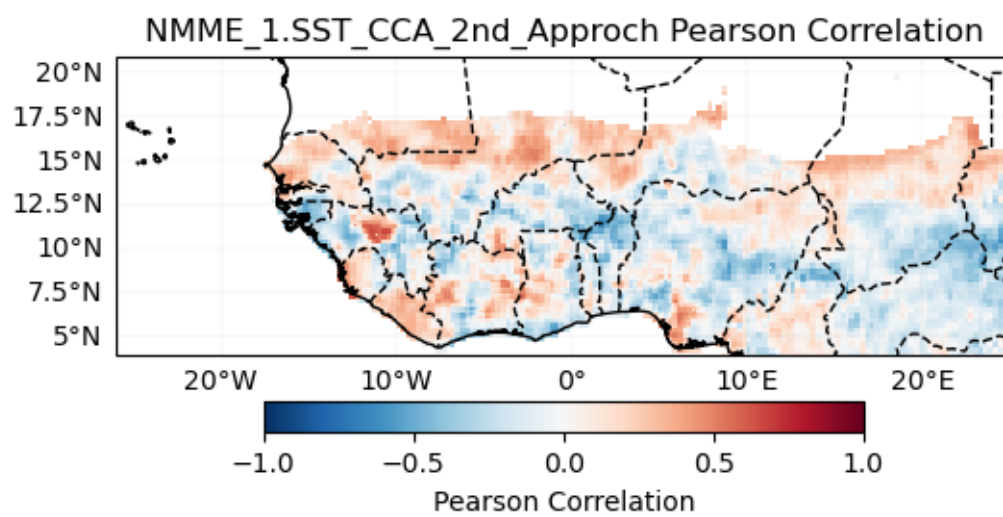
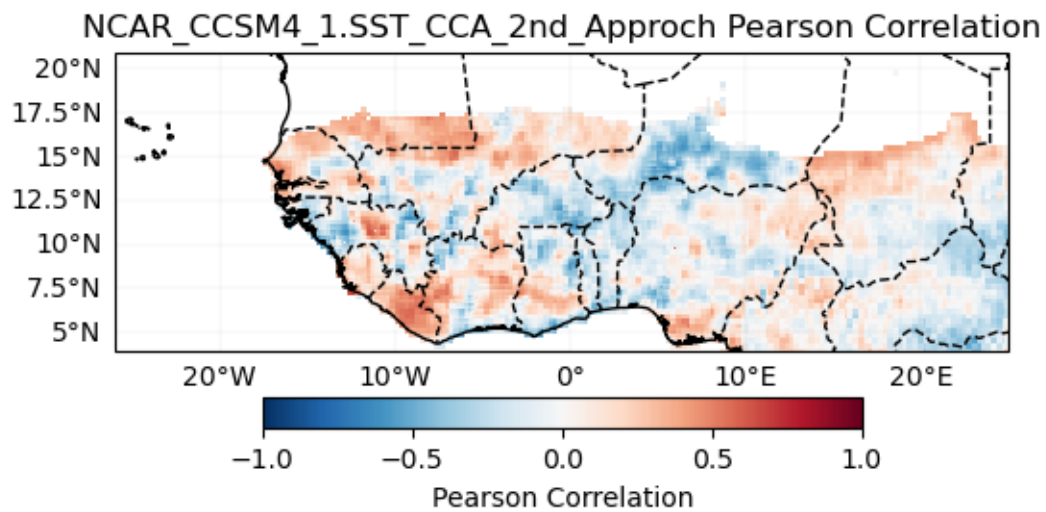




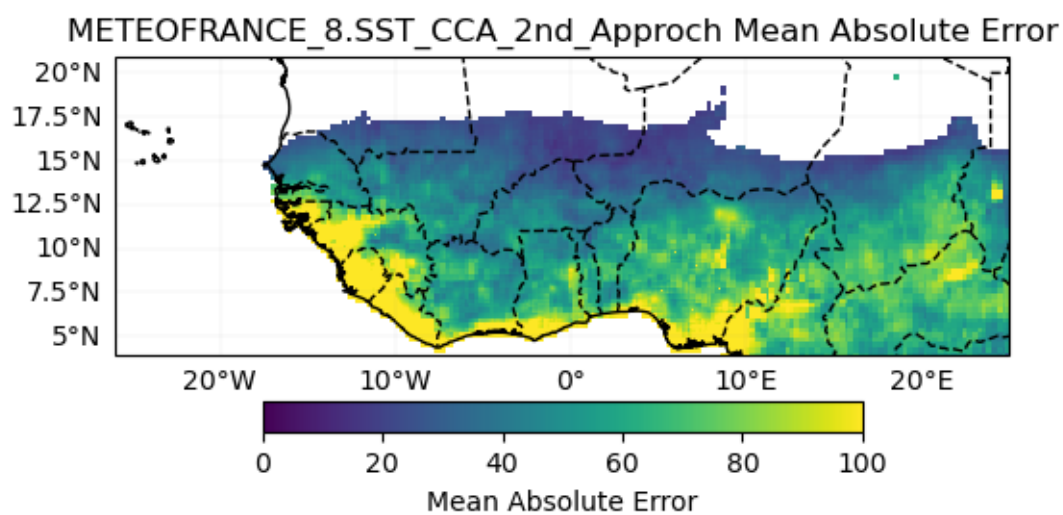
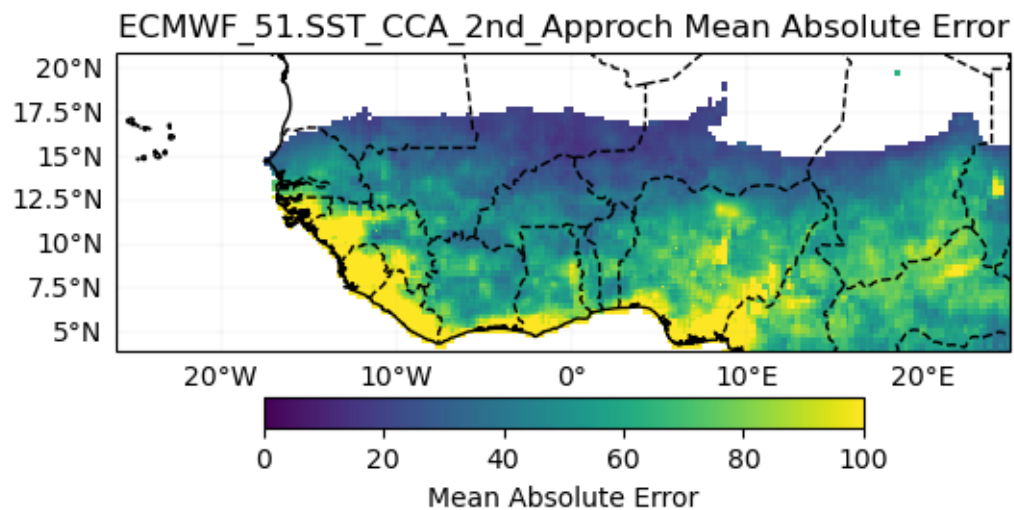


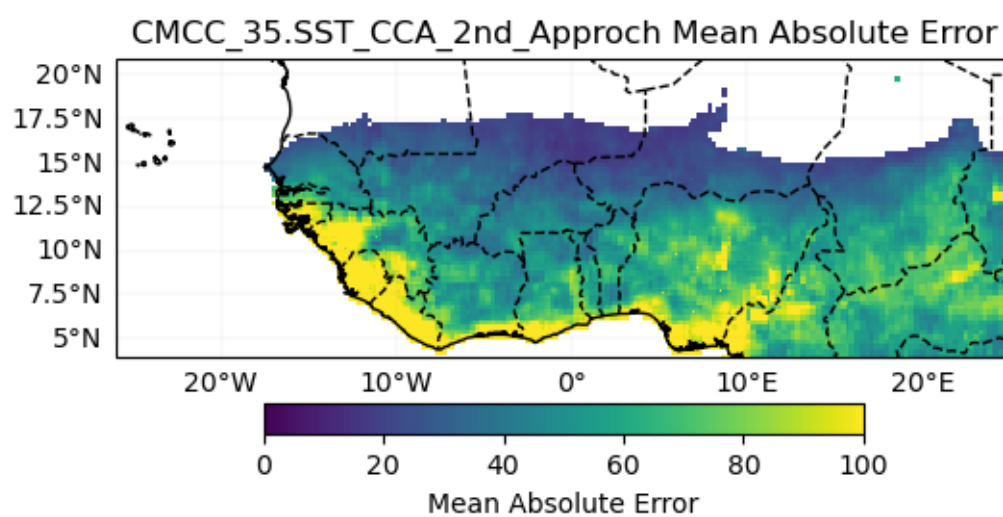
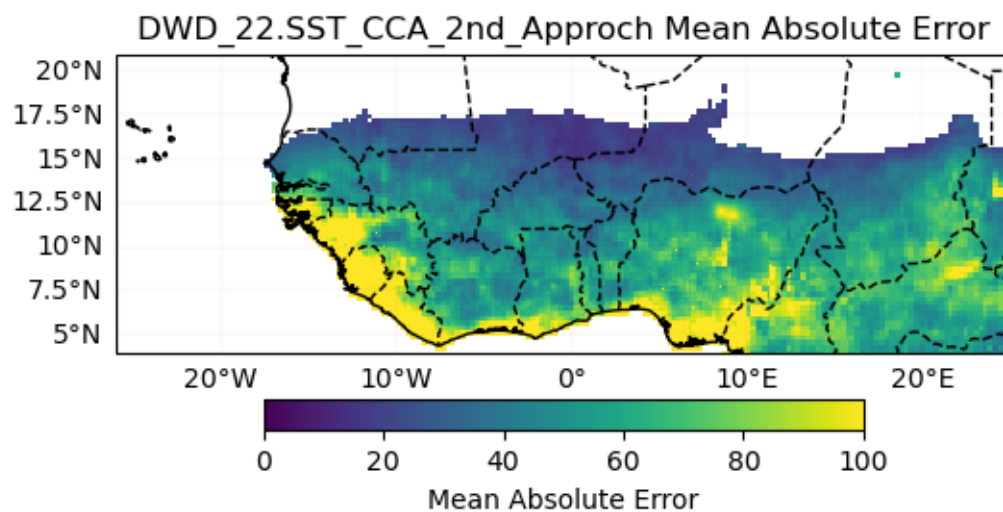


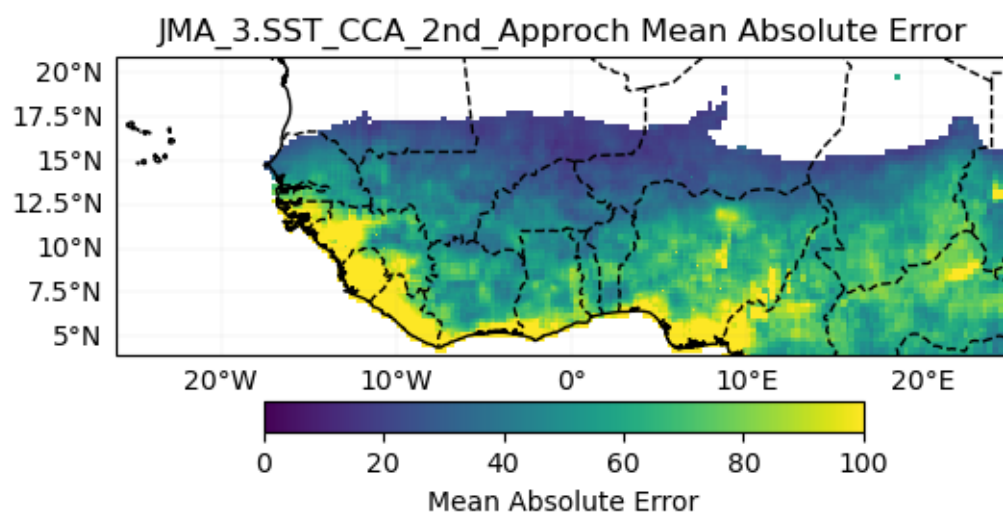
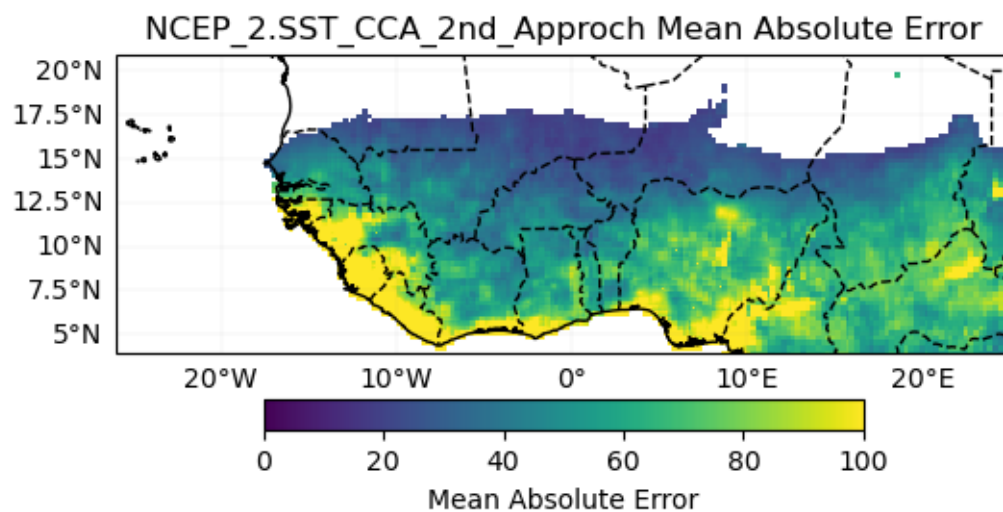


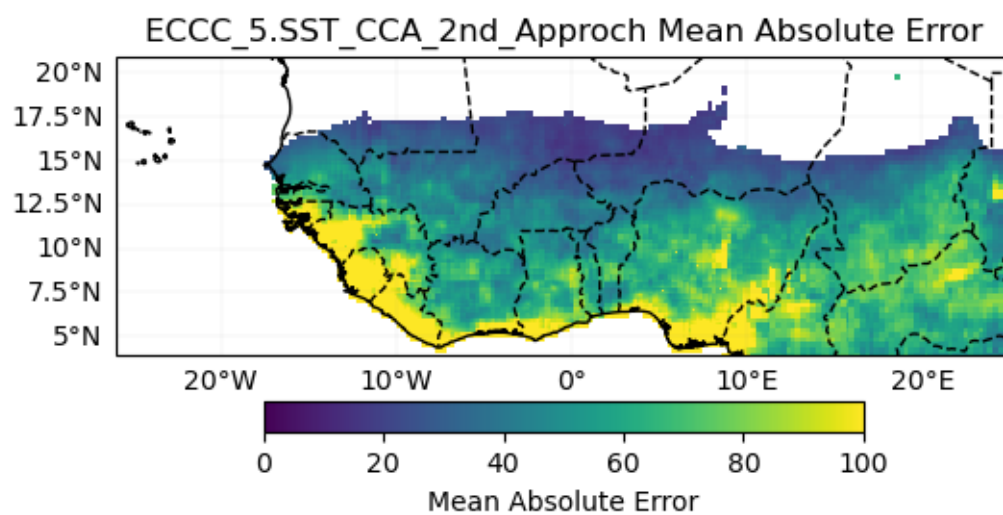
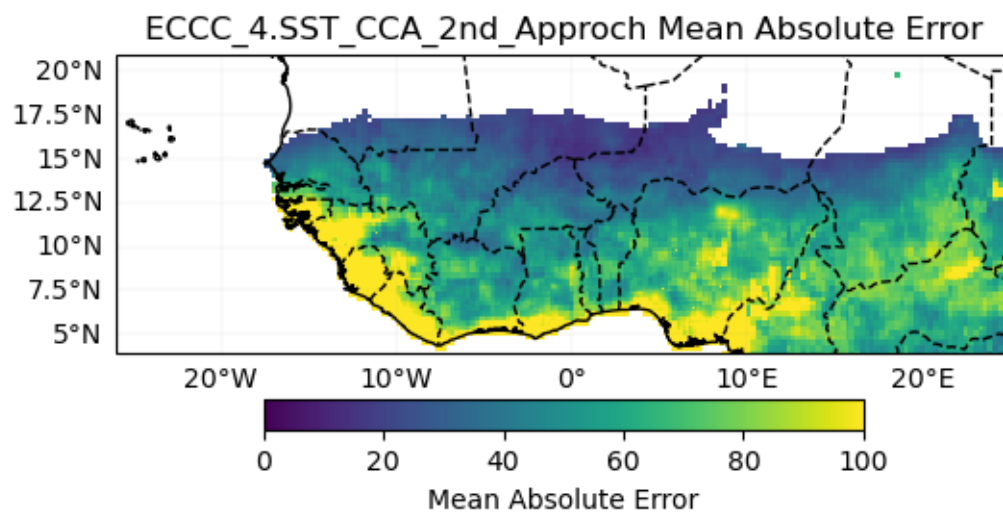


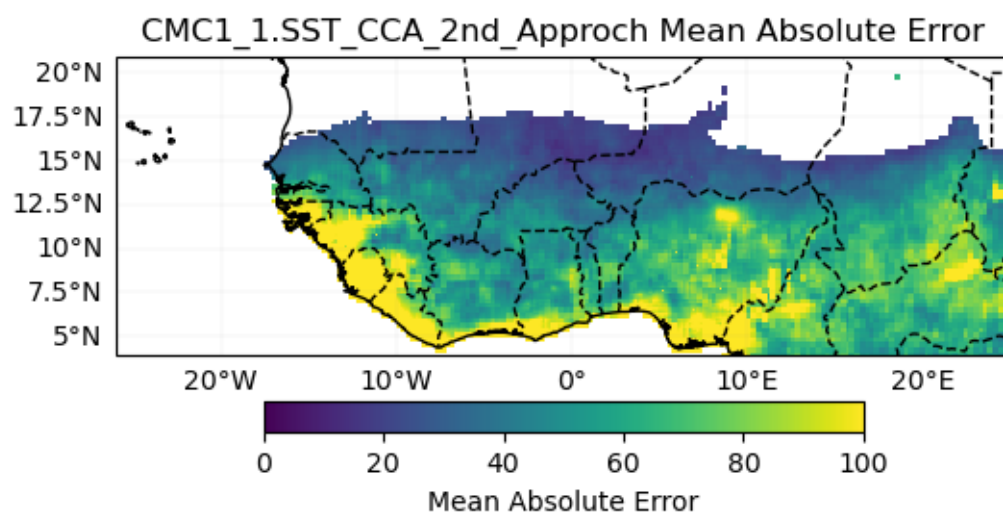
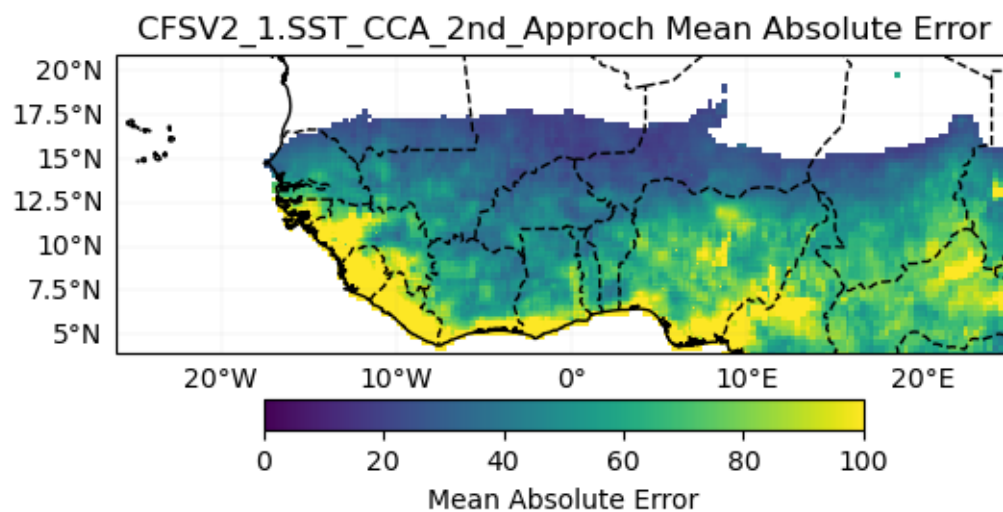
Mean Absolute Error

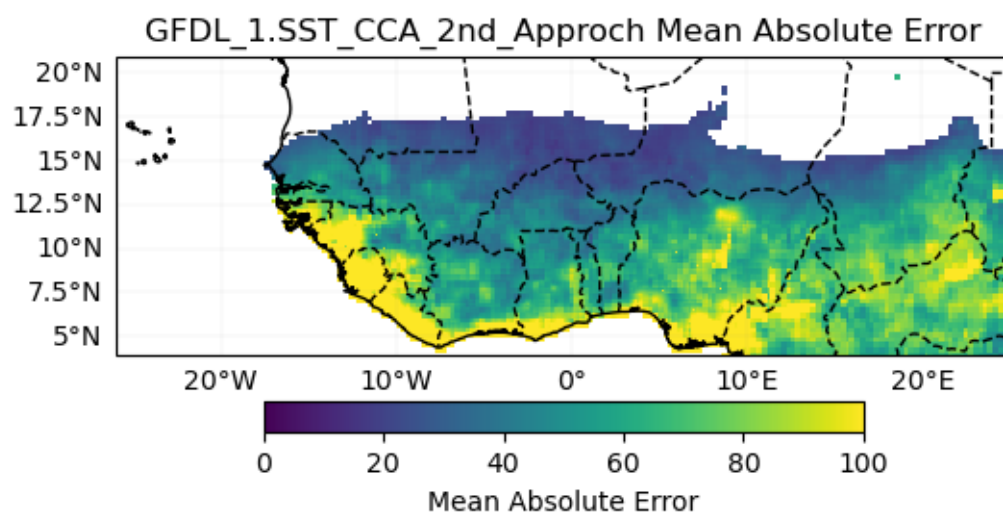
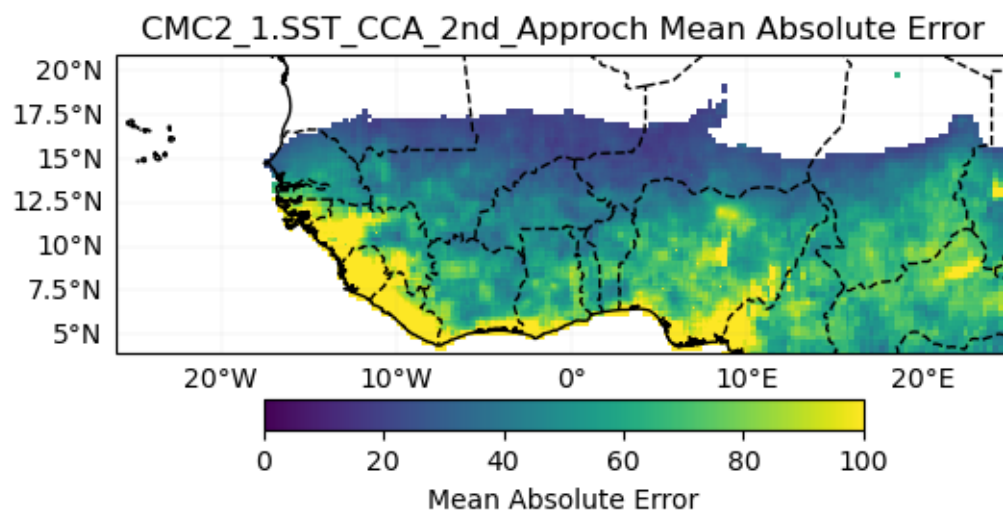


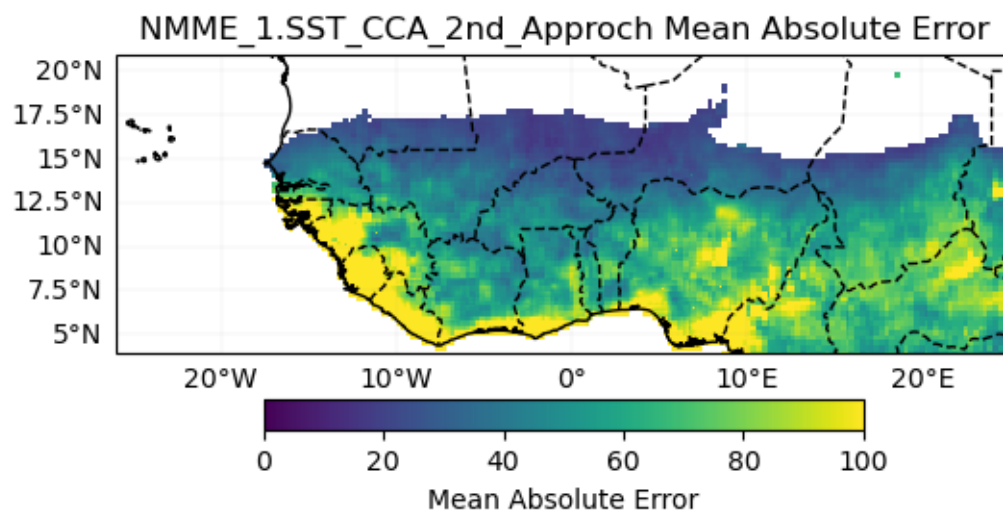
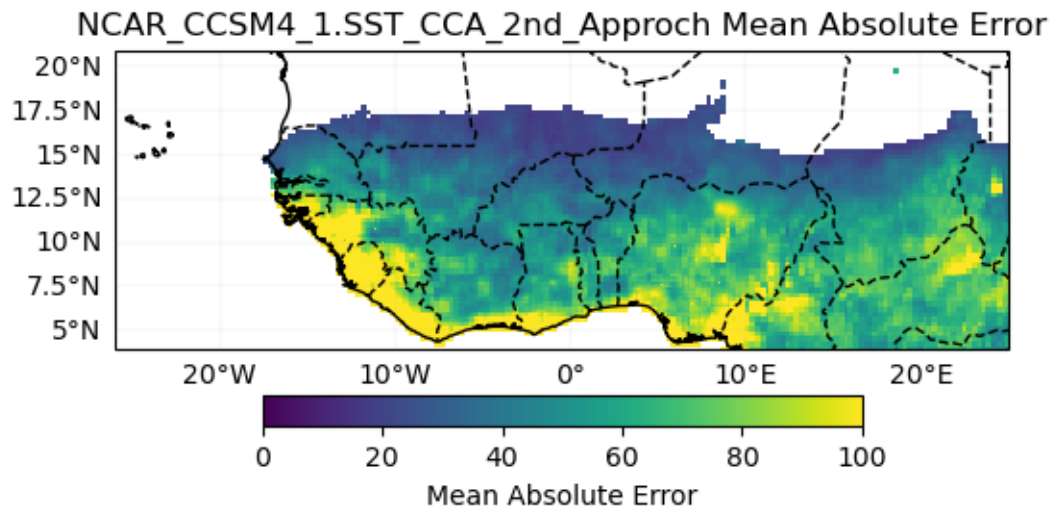








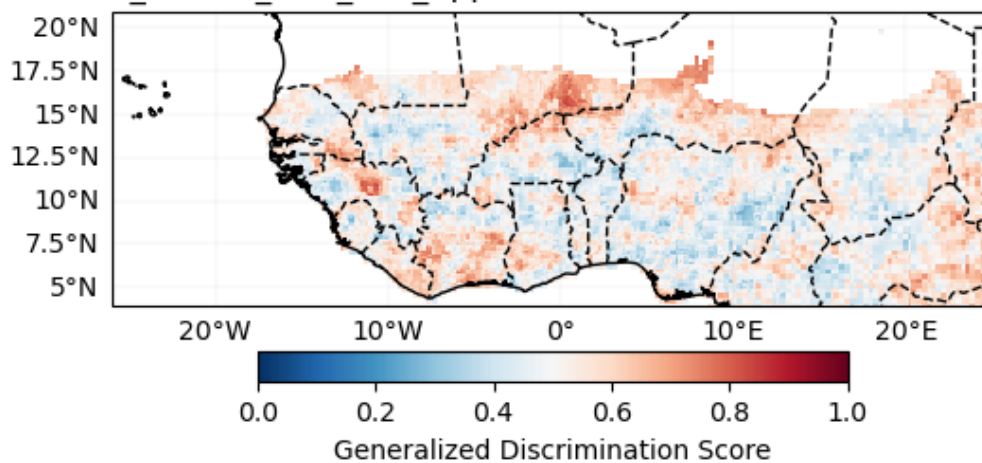




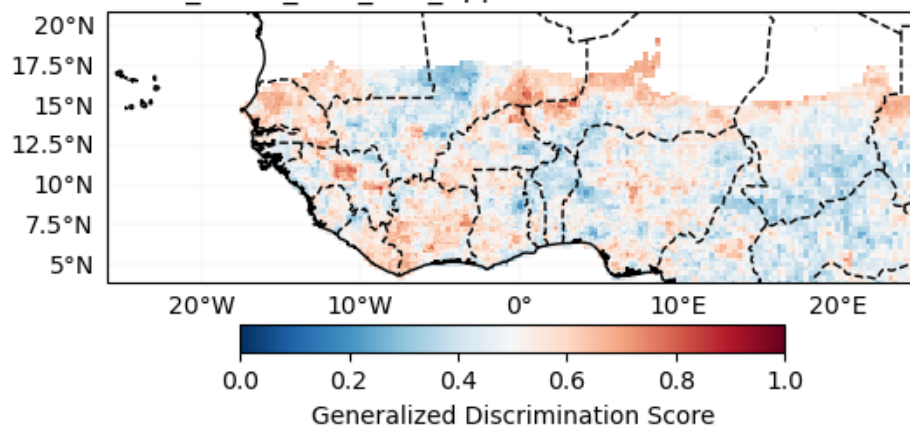
Probabilistic

Generalized Discrimination Score

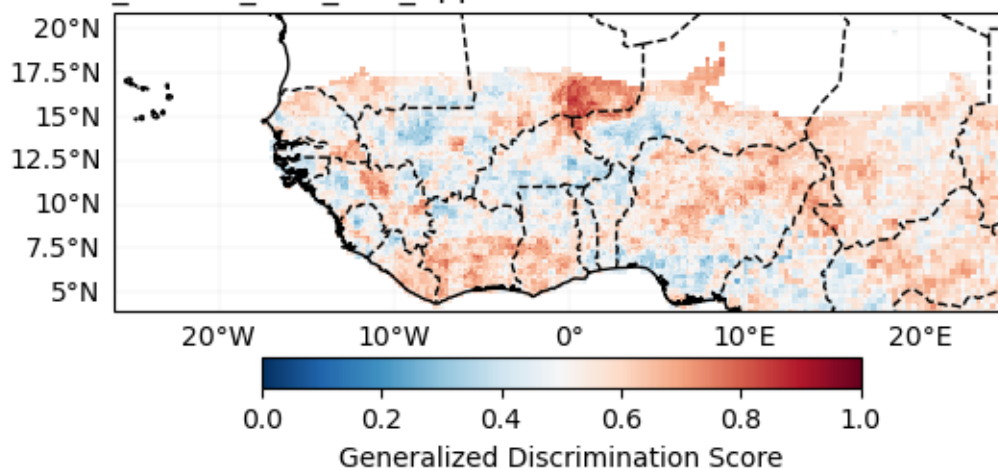
ECMWF_51.SST_CCA_2nd_Approch Generalized Discrimination Score



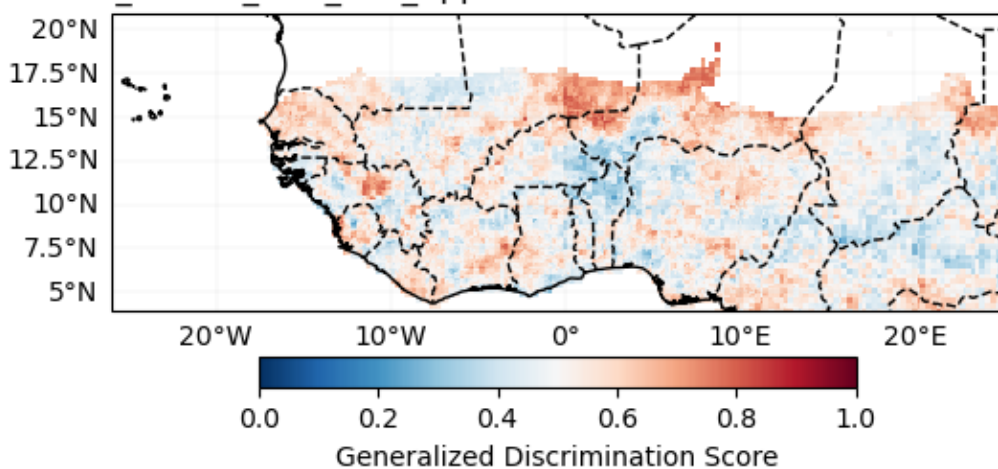
METEOFRACTANCE_8.SST_CCA_2nd_Approch Generalized Discrimination Score



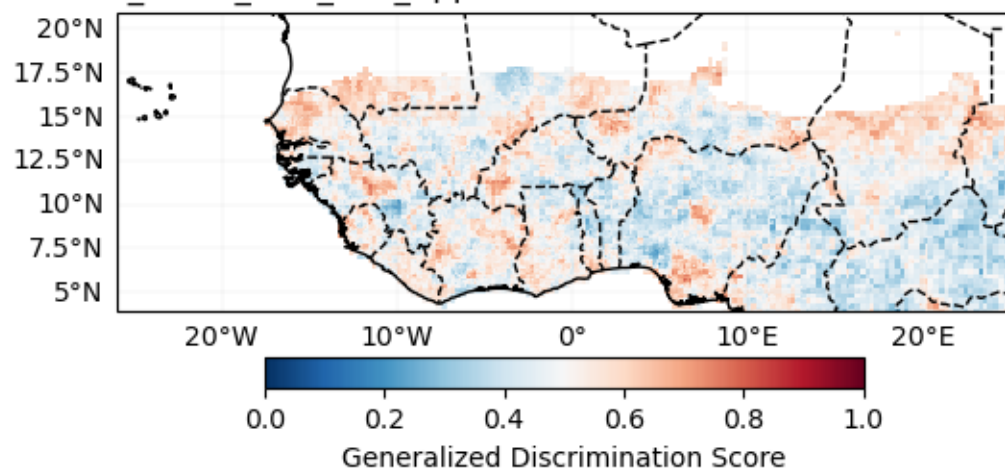
DWD_22.SST_CCA_2nd_Approch Generalized Discrimination Score



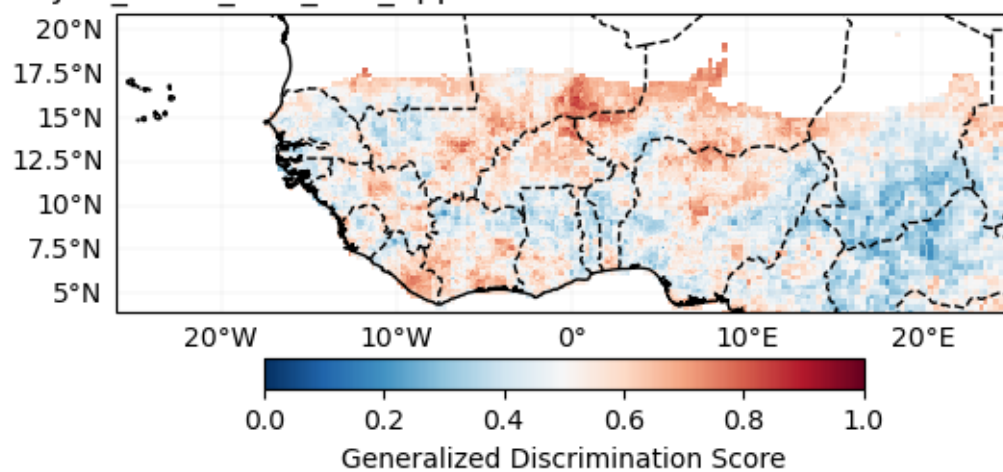
CMCC_35.SST_CCA_2nd_Approch Generalized Discrimination Score



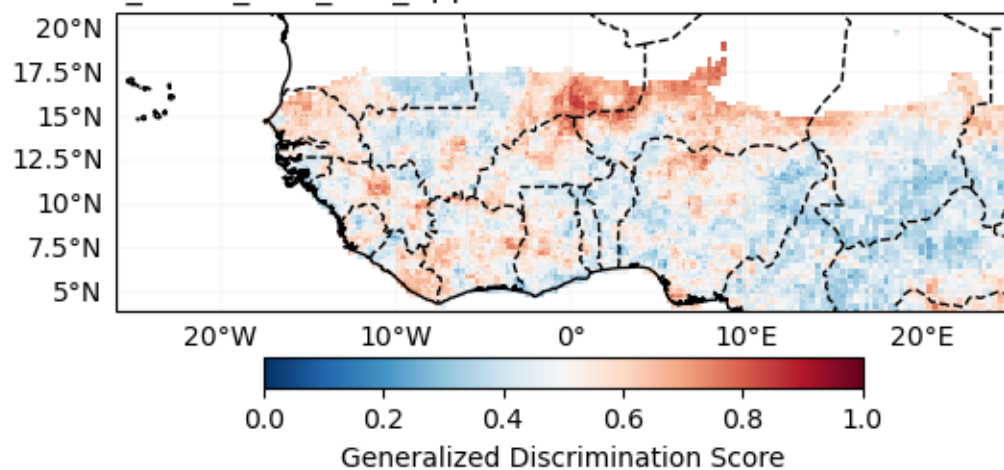
NCEP_2.SST_CCA_2nd_Approch Generalized Discrimination Score



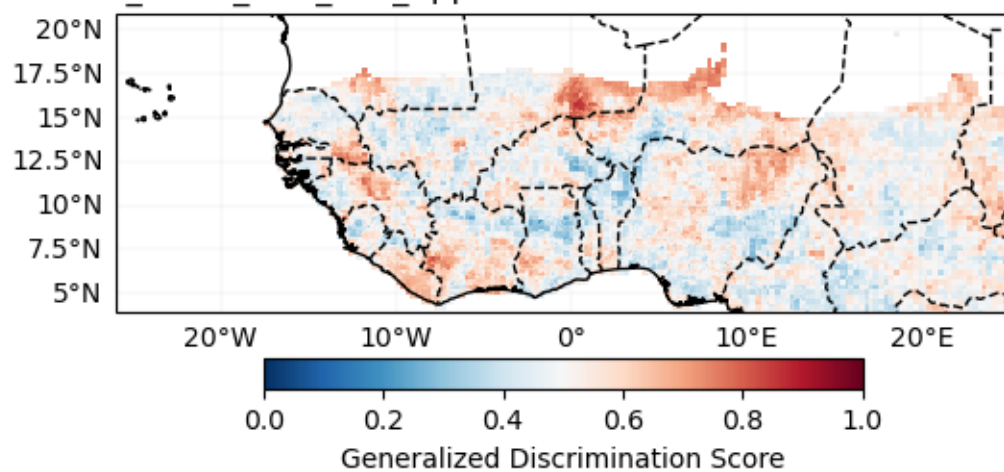
JMA_3.SST_CCA_2nd_Approch Generalized Discrimination Score



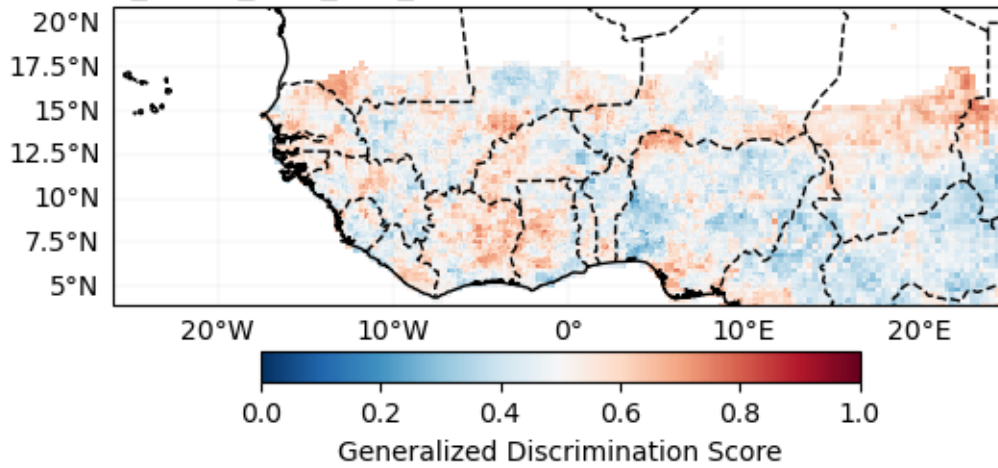
ECCC_4.SST_CCA_2nd_Approch Generalized Discrimination Score



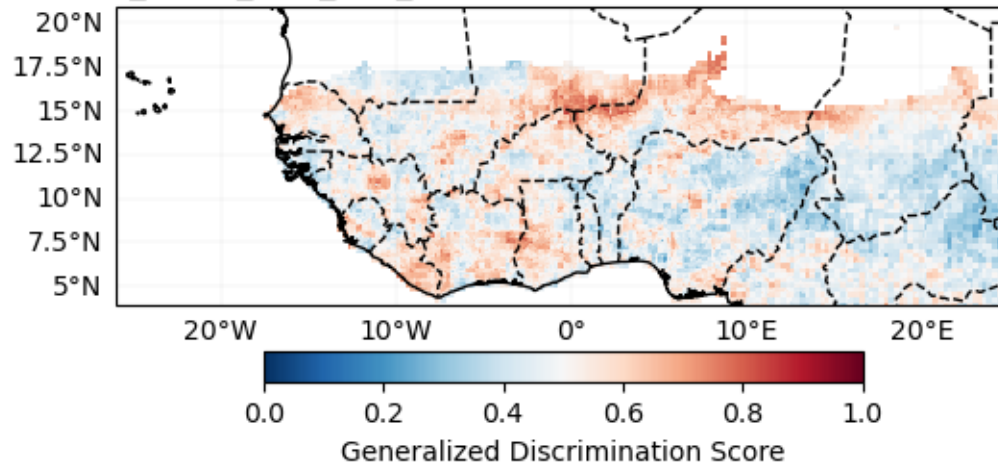
ECCC_5.SST_CCA_2nd_Approch Generalized Discrimination Score



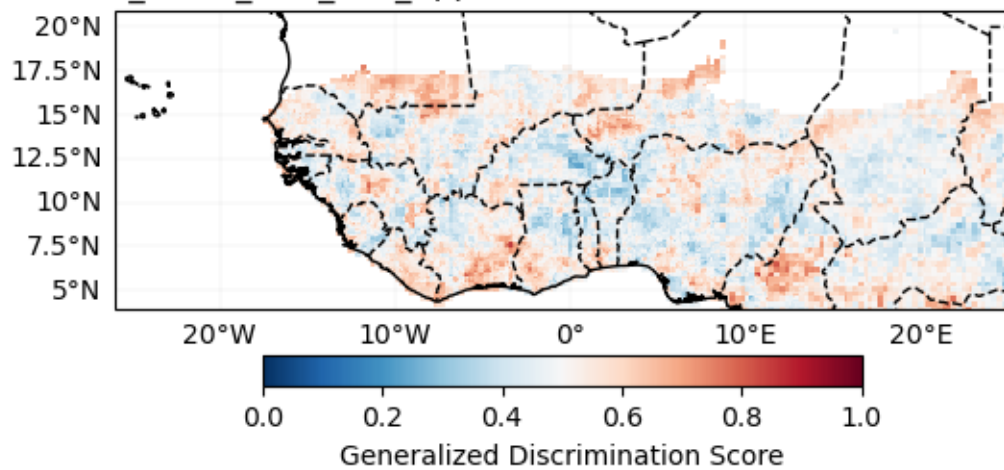
CFSV2_1.SST_CCA_2nd_Approch Generalized Discrimination Score



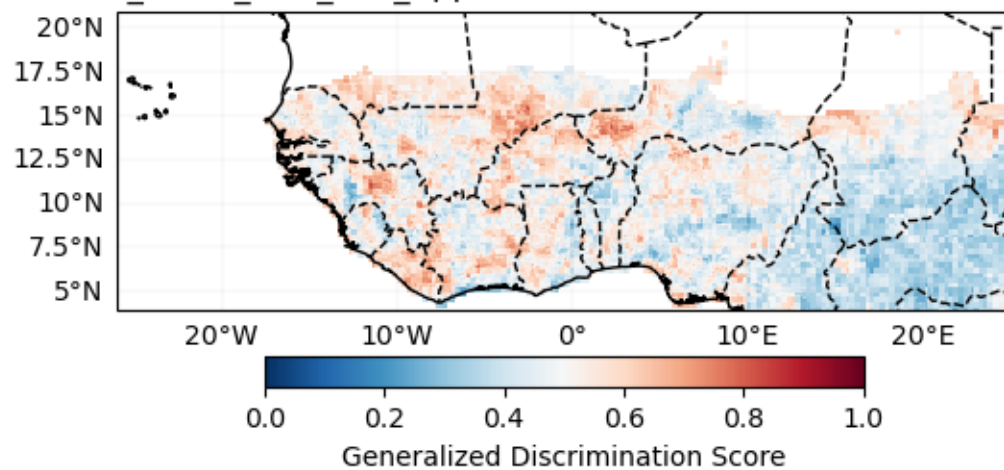
CMC1_1.SST_CCA_2nd_Approch Generalized Discrimination Score



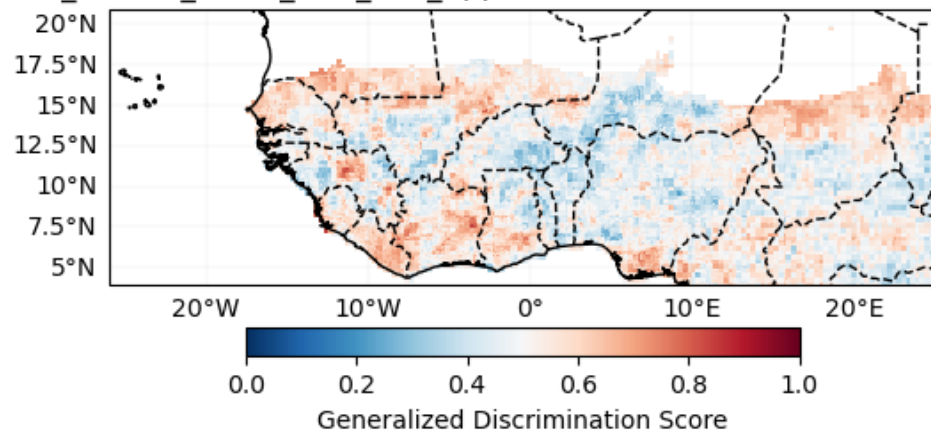
CMC2_1.SST_CCA_2nd_Approch Generalized Discrimination Score



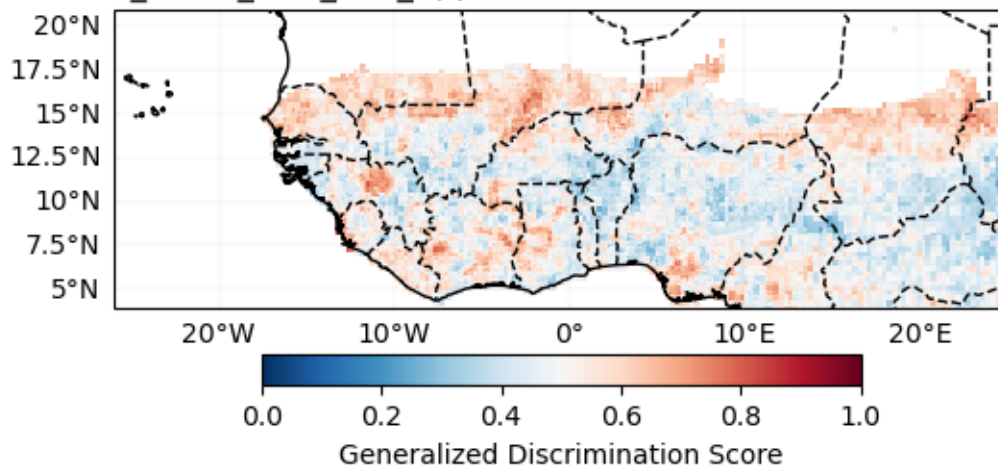
GFDL_1.SST_CCA_2nd_Approch Generalized Discrimination Score



NCAR_CCSM4_1.SST_CCA_2nd_Approch Generalized Discrimination Score



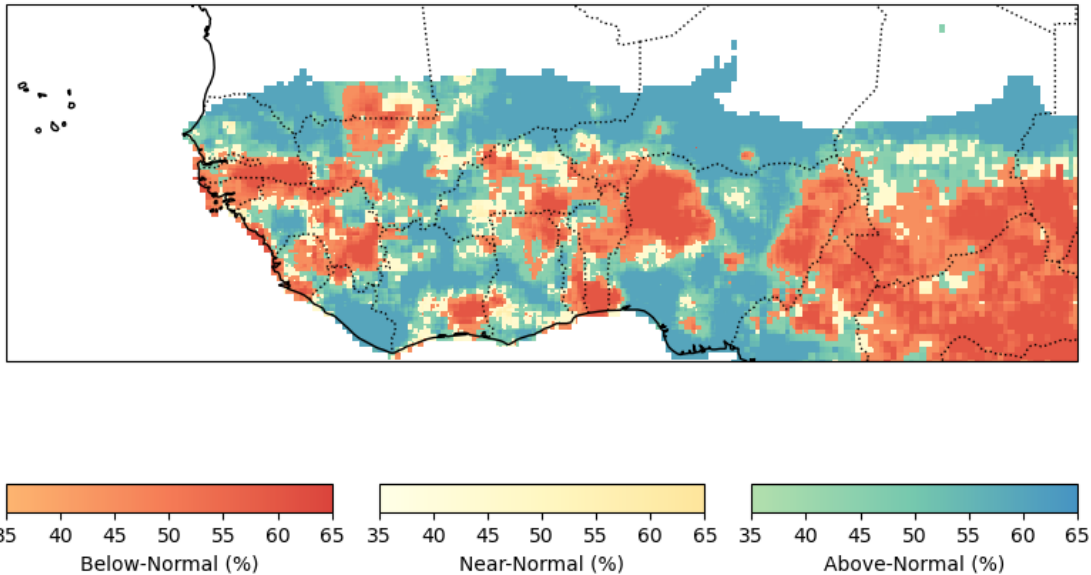
NMME_1.SST_CCA_2nd_Approch Generalized Discrimination Score



Forecast (individual)

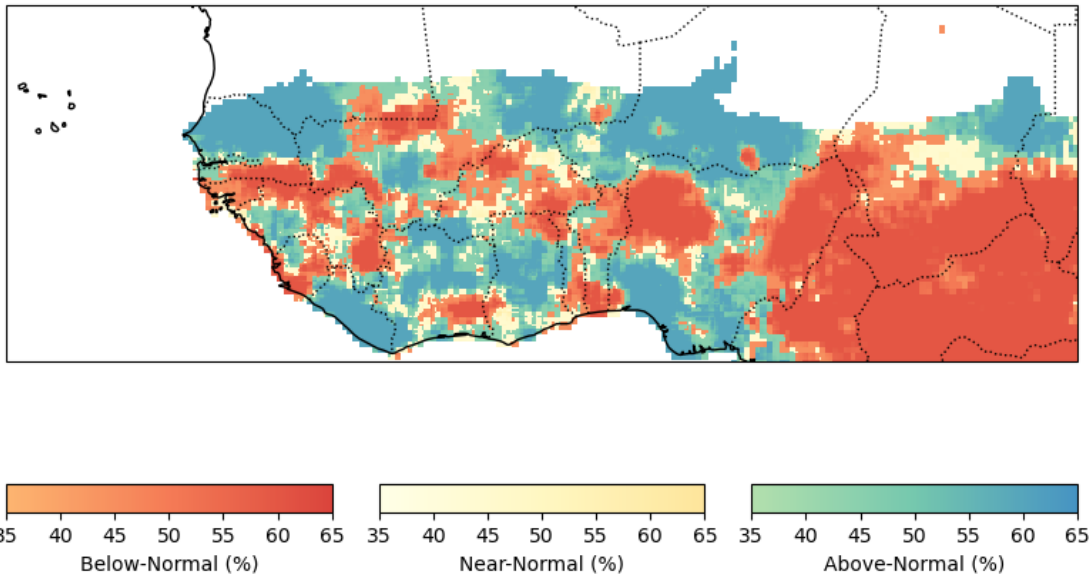
ECMWF_51

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



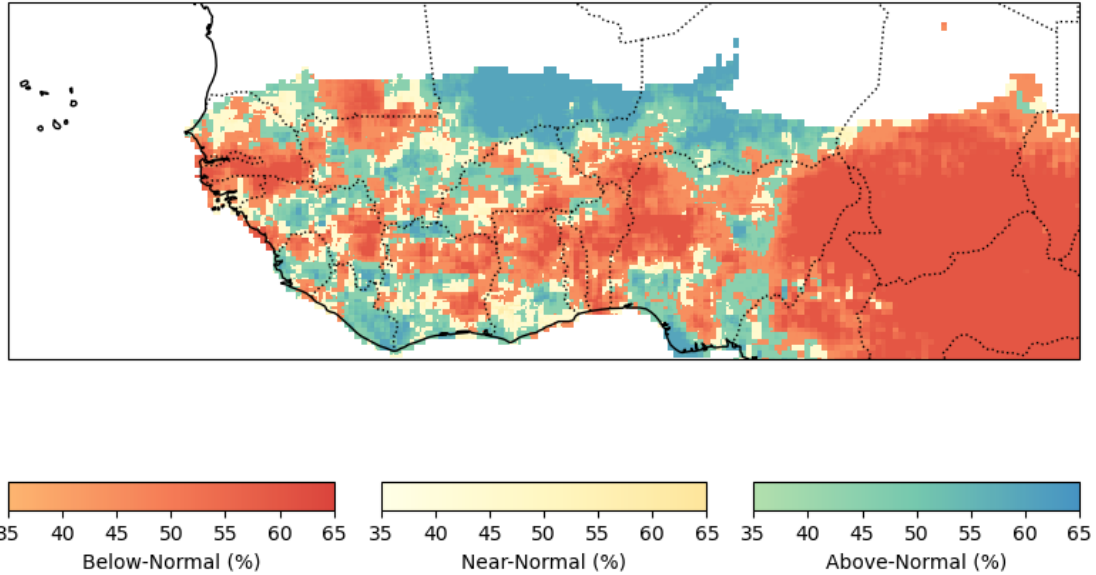
METEOFRANCE_8

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



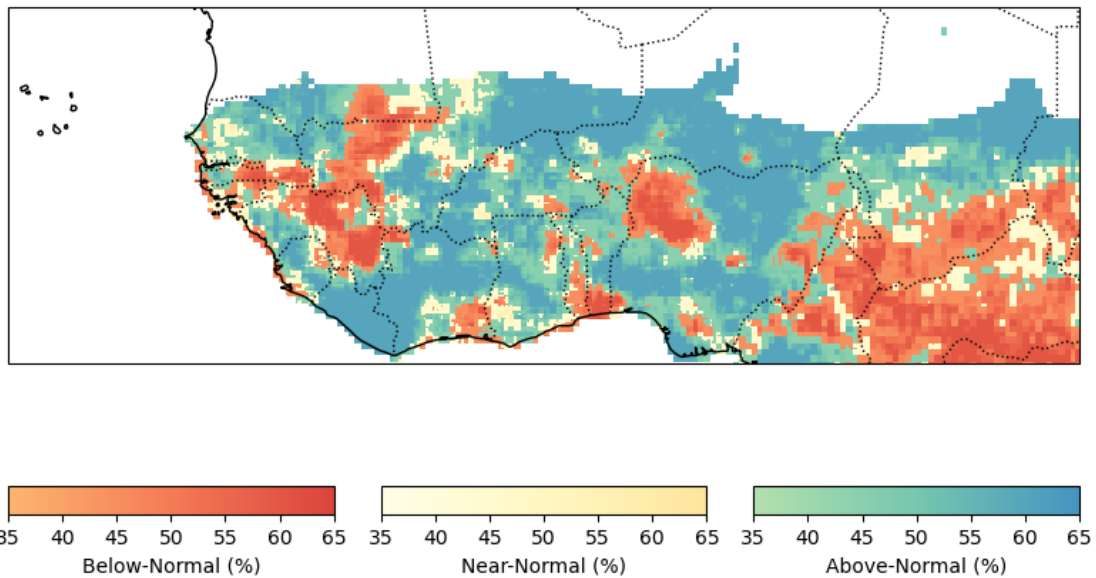
DWD_22

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



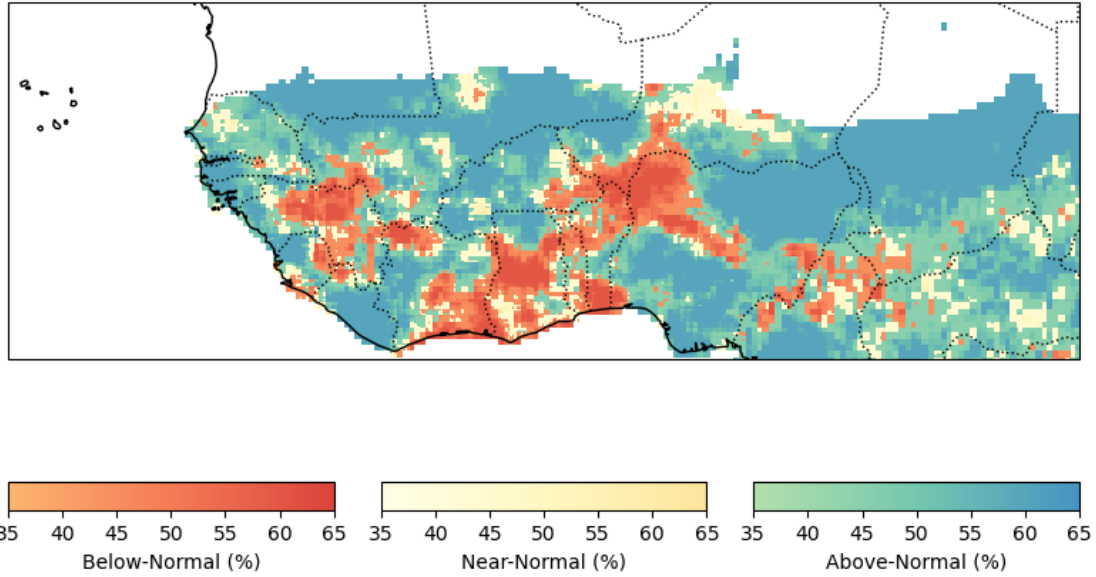
CMCC_35

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



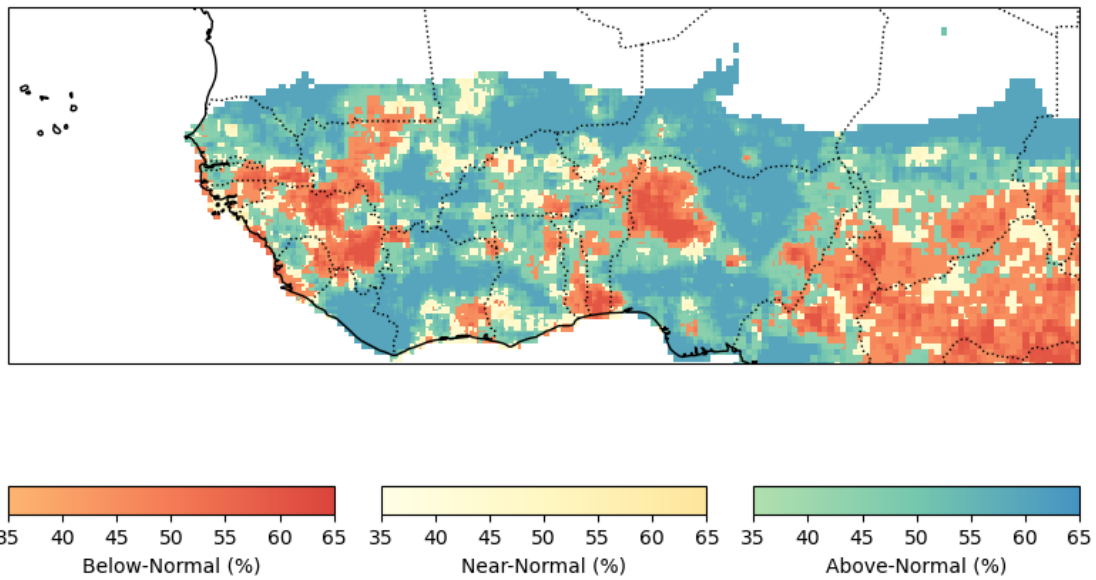
NCEP_2

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



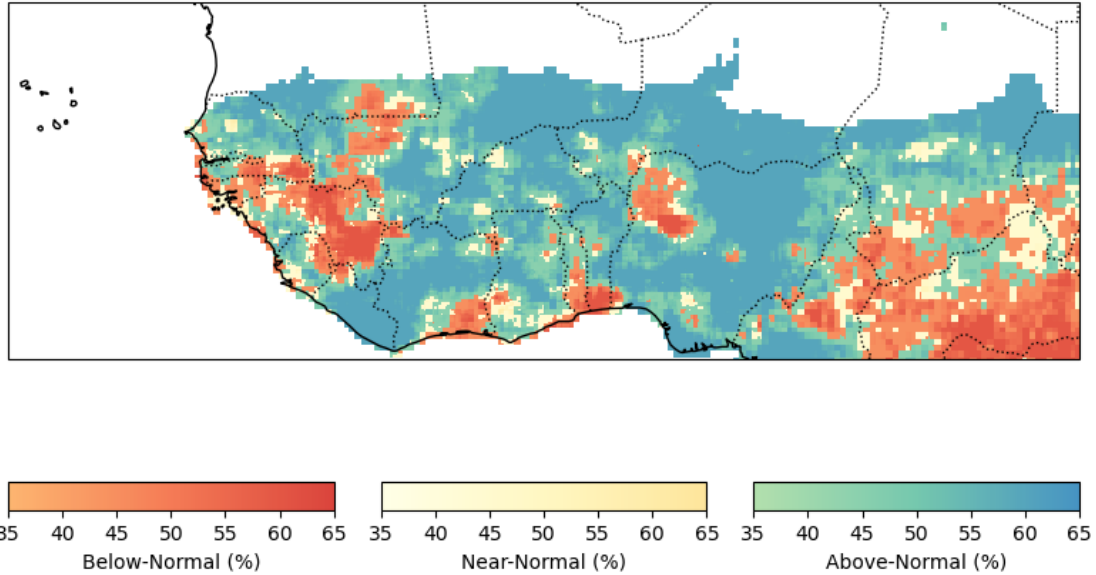
JMA_3

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



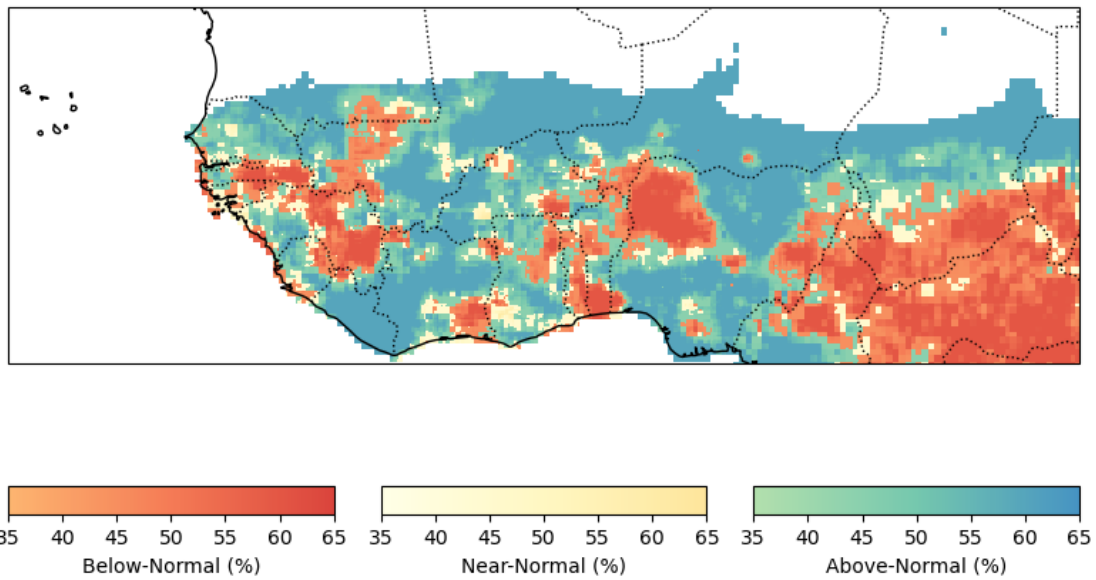
ECCC_4

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



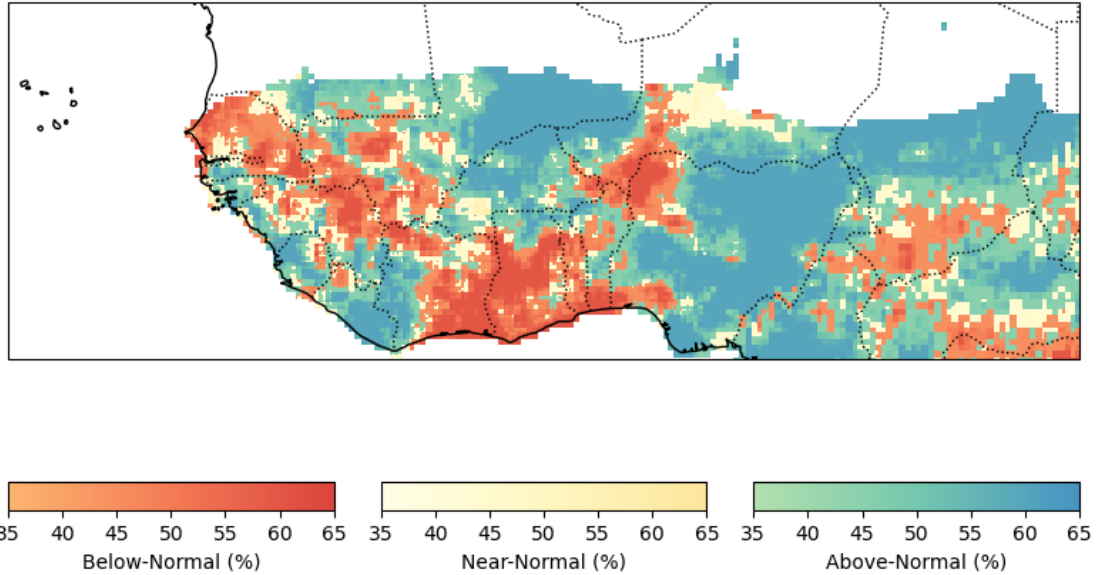
ECCC_5

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



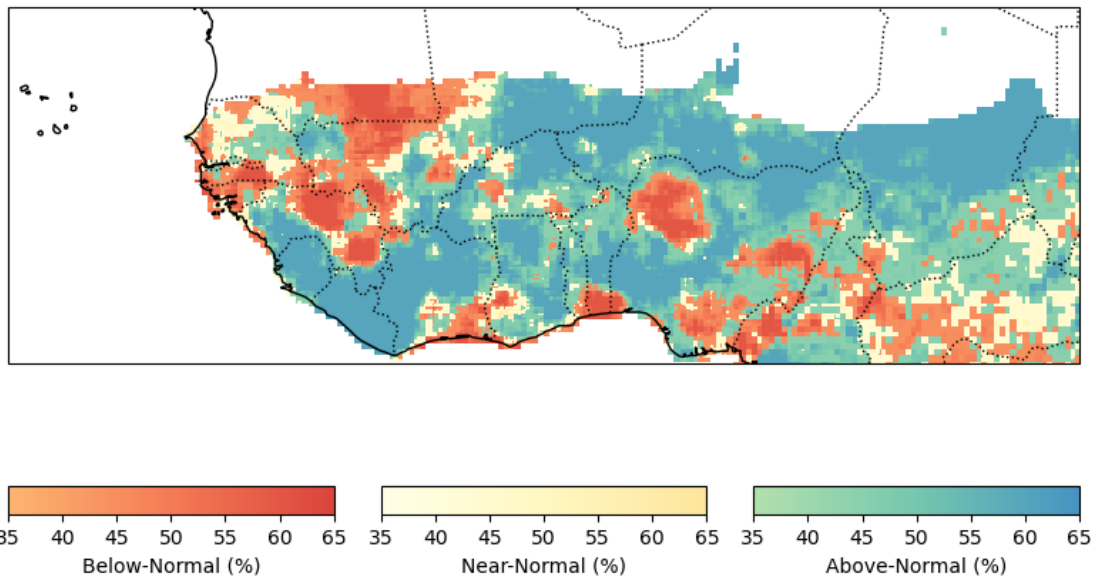
CFSV2_1

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



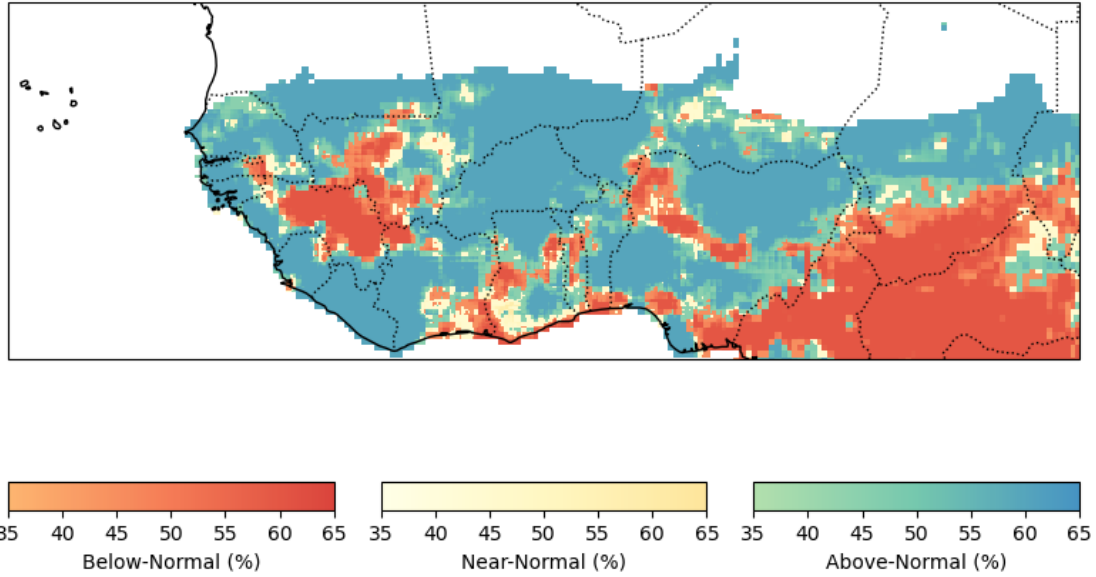
CMC1_1

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



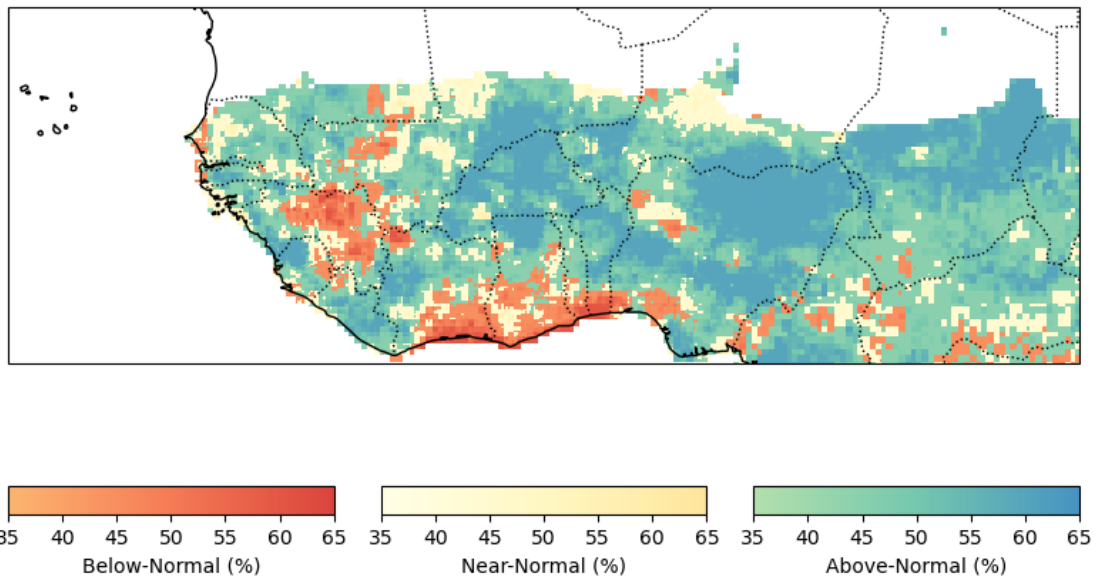
CMC2_1

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



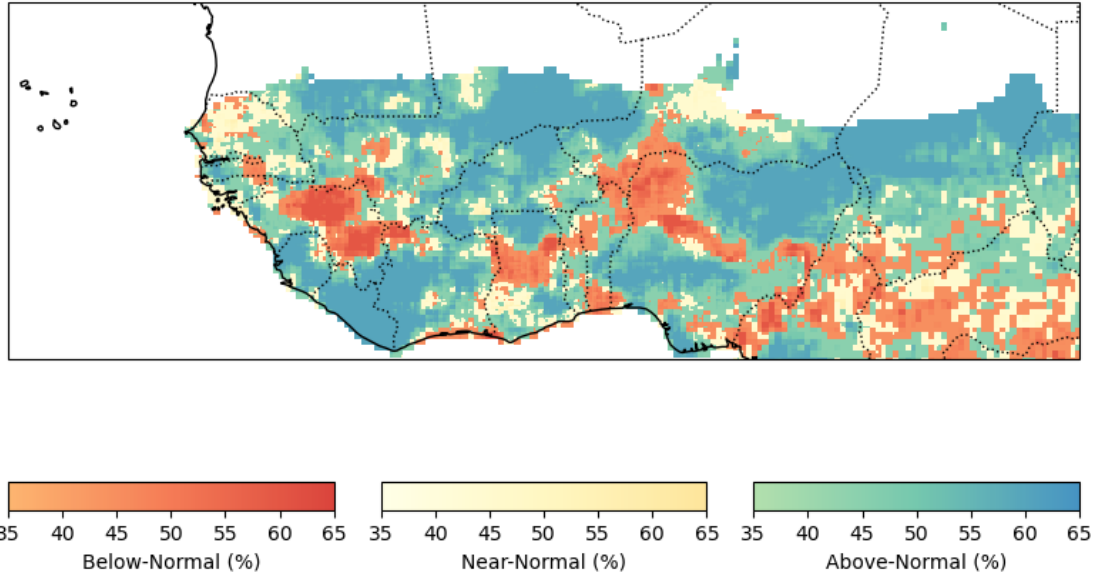
GFDL_1

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



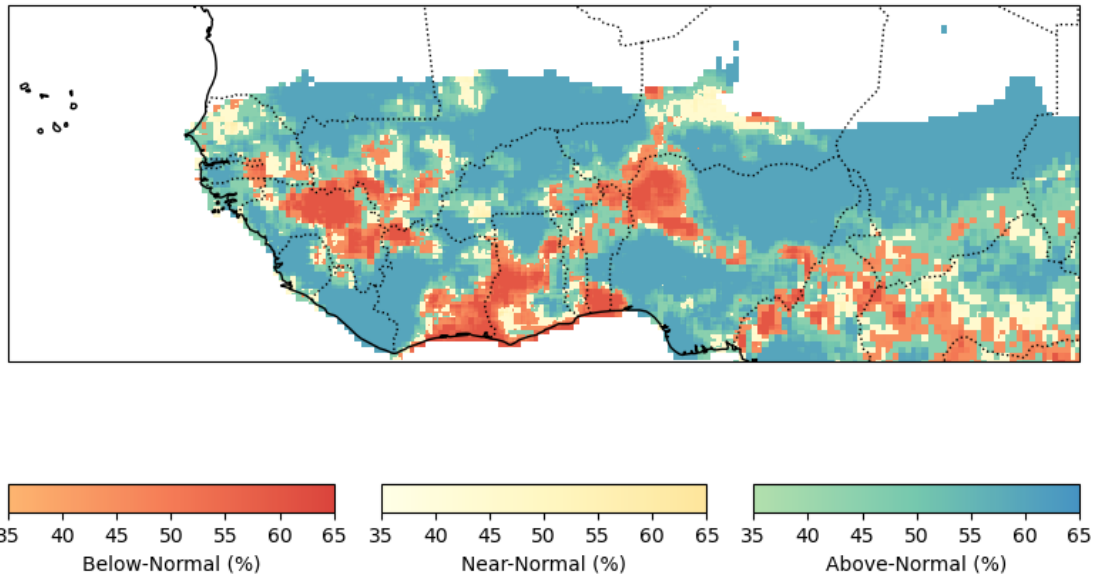
NCAR_CCSM4_1

Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



NMME_1

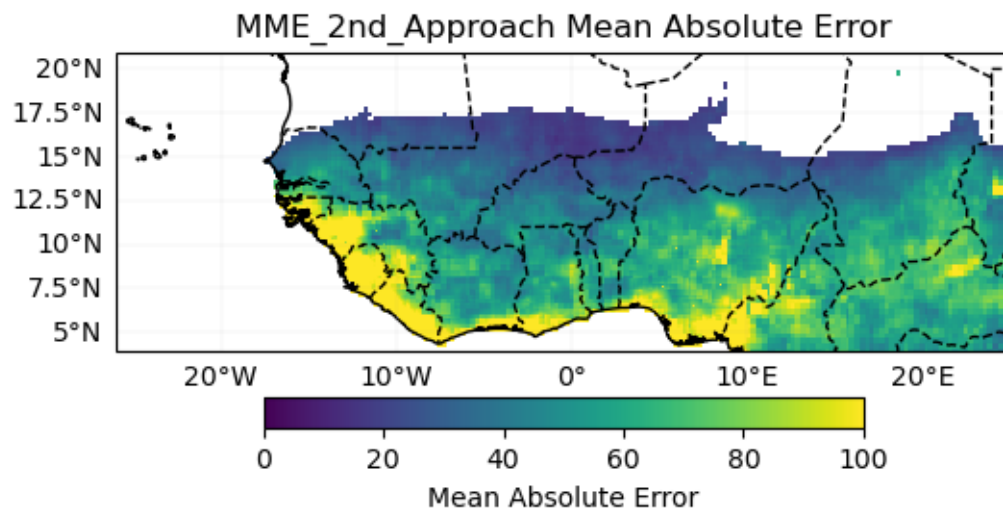
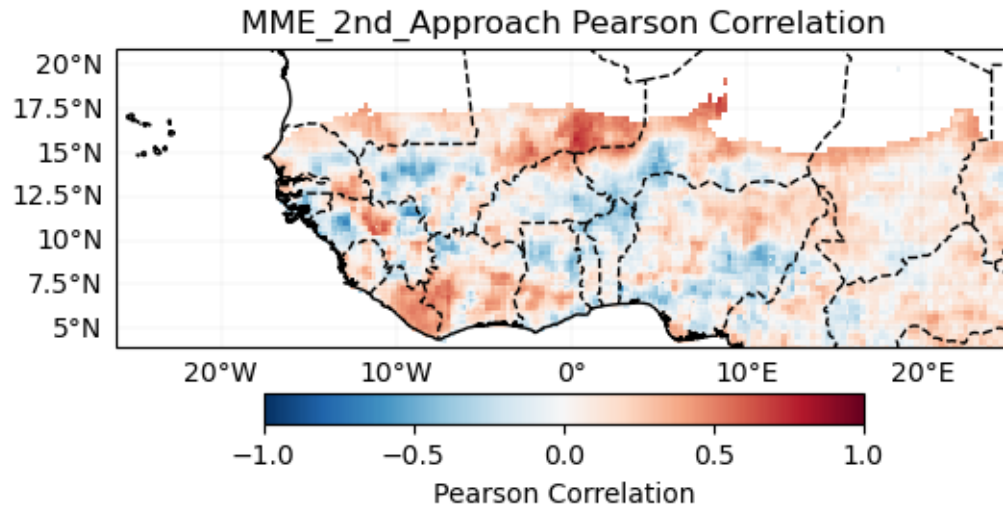
Probabilistic Forecast - CCA_2nd_Approch MayJunJul-2025



2.0.2 MME 2nd-Approach: Statistico-Dynamical

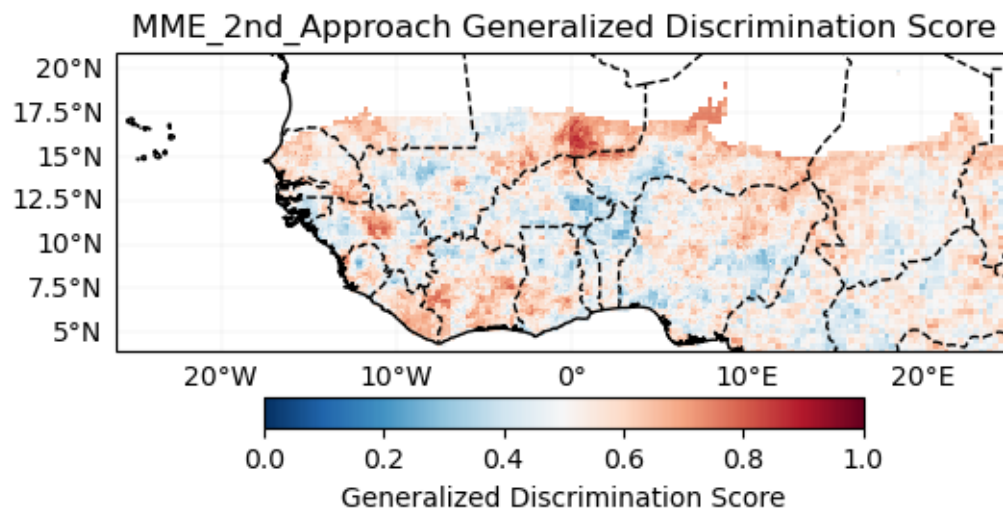
['DWD_22.SST', 'ECMWF_51.SST', 'ECCC_5.SST', 'CMCC_35.SST', 'NCAR_CCSM4_1.SST']

Verification Deterministic

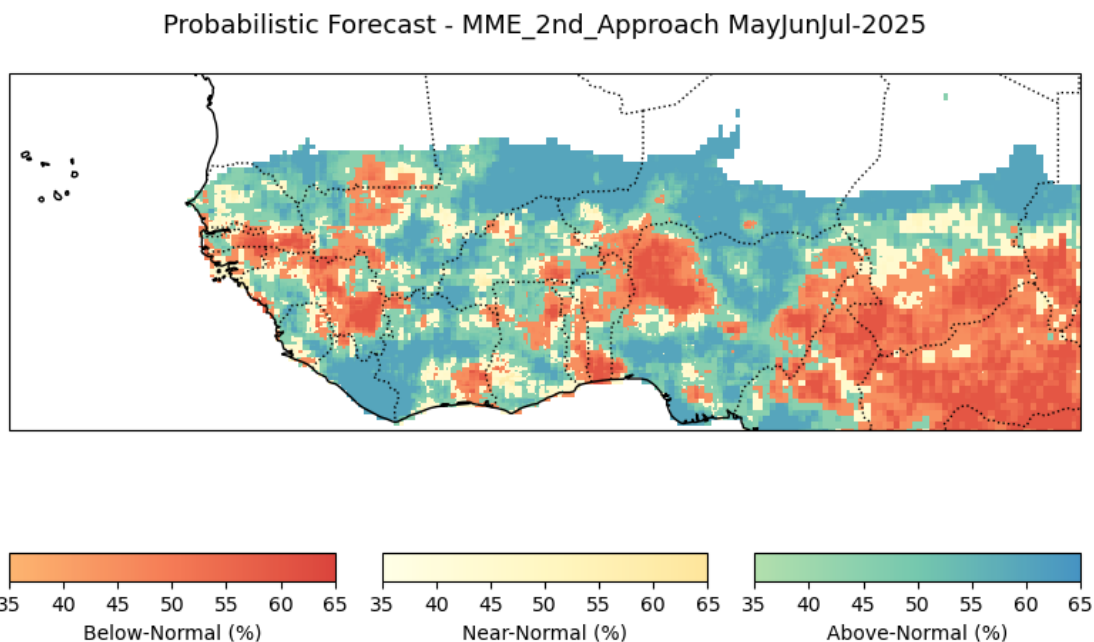


Probabilistic

Generalized Discrimination Score



Forecast



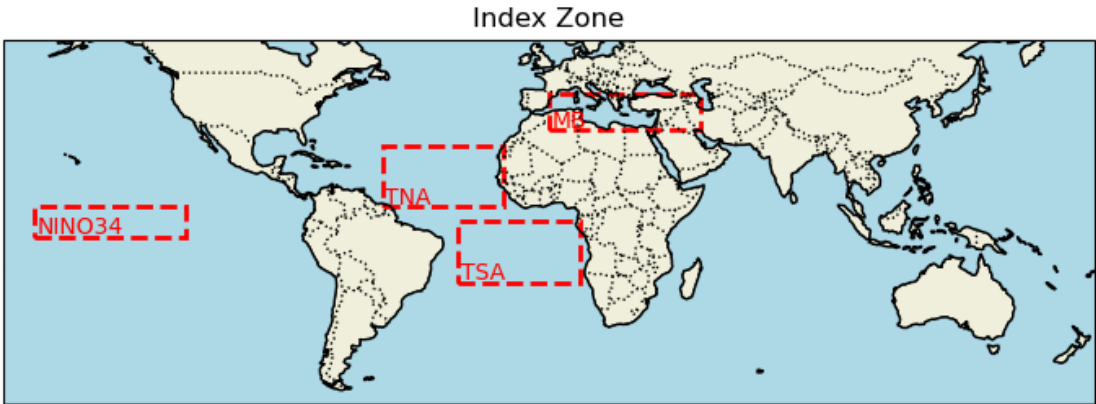
42755

3 3rd - Approach: Observational-based seasonal forecast

MJJ_2025/reanalysis_data/NOAA_SST_1991_2025_JanFebMar.nc exists. Skipping.

Process SST index

```
['NINO34', 'NINO12', 'NINO3', 'NINO4', 'NINO_Global', 'TNA', 'TSA', 'NAT',  
'SAT', 'TASI', 'WTIO', 'SETIO', 'DMI', 'MB']
```



NOAA SST

Calculate variance inflation factor to see multicollinearity predictor

	feature	VIF
0	NINO34	1.268913
1	TNA	1.698945
2	TSA	2.233701
3	DMI	1.298347
4	MB	1.718974

```
['NINO34', 'TNA', 'TSA', 'DMI', 'MB']
```

3.1 Initialize WAS_LinearRegression_Model

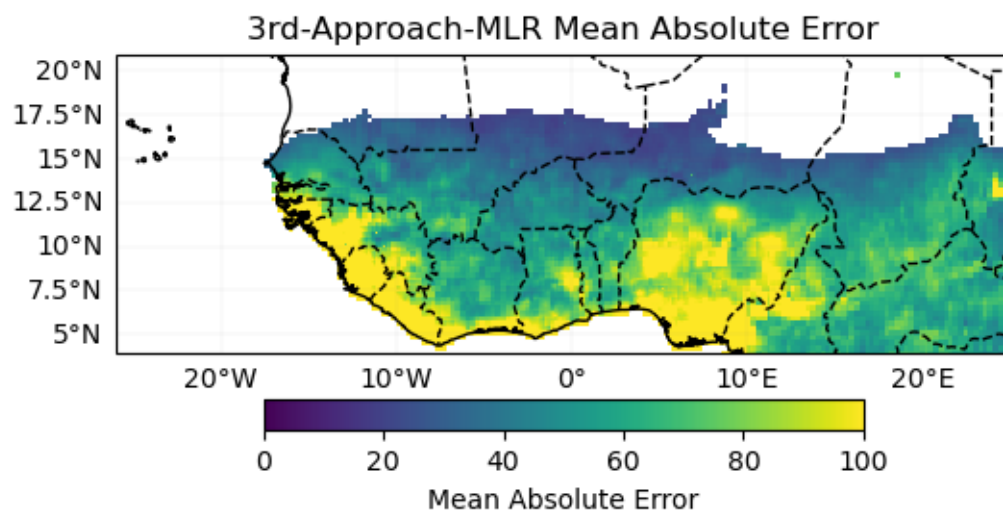
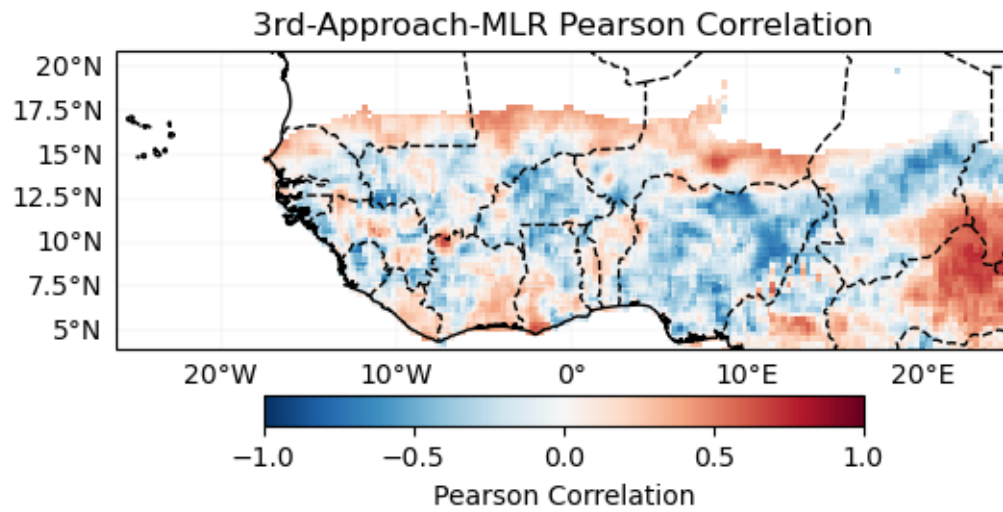
8

Cross validation

Cross-validation ongoing

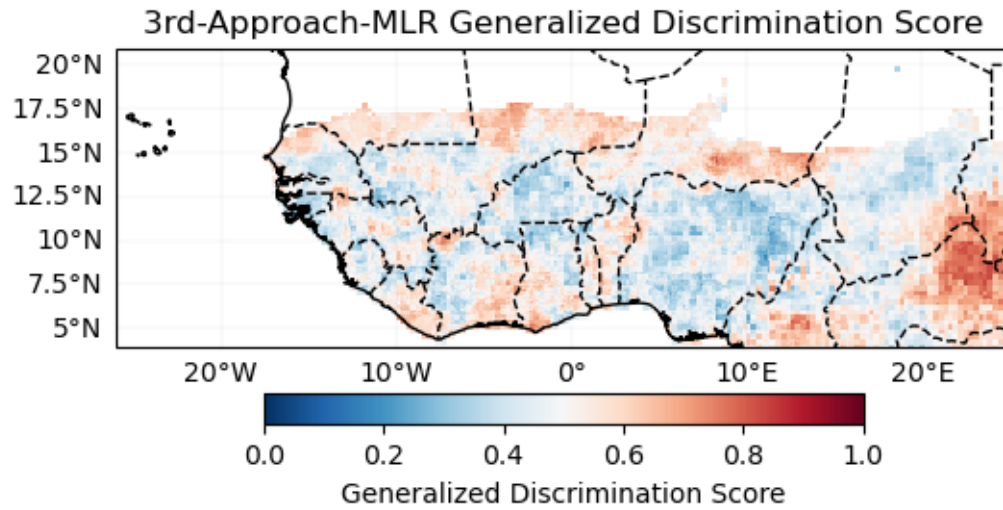
100%| | 34/34 [08:13<00:00, 14.52s/it]

Verification Deterministic

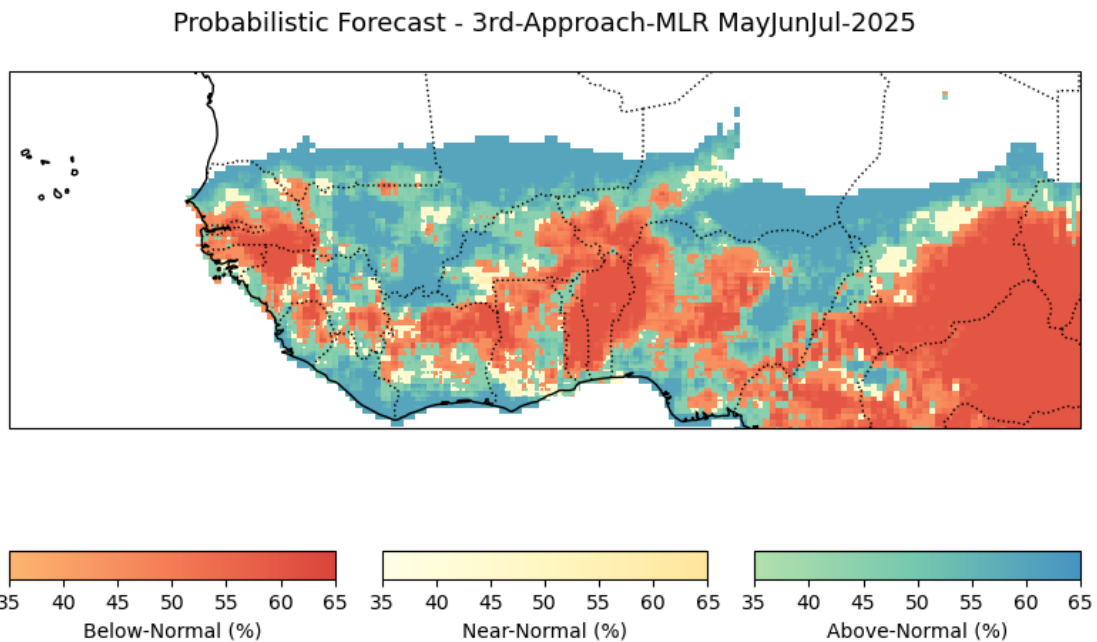


Probabilistic

Generalized Discrimination Score



Forecast



4 4th - Analog-based seasonal forecast

4.1 Initialize WAS_Analog class

Cross-validation

MJJ_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
 already exists. Skipping download.
 Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
 month
 MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
 File 'MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
 exists. Skipping download.
 All requested SST data has been processed.
 Cross-validation ongoing

3%	1/34 [00:00<00:06, 5.36it/s]
similar years for 1991 are [2002 1997 2023]	
6%	2/34 [00:00<00:05, 5.34it/s]
similar years for 1992 are [2020 2019 2005]	
9%	3/34 [00:00<00:05, 5.26it/s]
similar years for 1993 are [2020 1991 1996]	
12%	4/34 [00:00<00:05, 5.27it/s]
similar years for 1994 are [2003 1991 2009]	
15%	5/34 [00:00<00:05, 5.30it/s]
similar years for 1995 are [2016 1998 2020]	
18%	6/34 [00:01<00:05, 5.33it/s]
similar years for 1996 are [1993 2018 2002]	
similar years for 1997 are [2023 2015 2009]	
24%	8/34 [00:01<00:04, 5.26it/s]
similar years for 1998 are [2010 2016 2024]	
26%	9/34 [00:01<00:04, 5.29it/s]
similar years for 1999 are [2021 2011 2008]	
29%	10/34 [00:01<00:04, 5.31it/s]
similar years for 2000 are [2012 2011 2009]	
32%	11/34 [00:02<00:04, 5.34it/s]
similar years for 2001 are [2023 2012 2017]	
35%	12/34 [00:02<00:04, 5.32it/s]
similar years for 2002 are [2023 1991 2018]	
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similar years for 2003 are [2024 1998 2016]	
41%	14/34 [00:02<00:03, 5.34it/s]

similar years for 2004 are [2007 2002 2018]
44%| | 15/34 [00:02<00:03, 5.33it/s]
similar years for 2005 are [2010 2016 2020]
47%| | 16/34 [00:03<00:03, 5.34it/s]
similar years for 2006 are [2009 2017 2011]
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similar years for 2007 are [2016 2010 2024]
53%| | 18/34 [00:03<00:02, 5.34it/s]
similar years for 2008 are [2021 2011 2018]
56%| | 19/34 [00:03<00:02, 5.35it/s]
similar years for 2009 are [2023 1997 2018]
59%| | 20/34 [00:03<00:02, 5.36it/s]
similar years for 2010 are [1998 2007 2016]
62%| | 21/34 [00:03<00:02, 5.37it/s]
similar years for 2011 are [2008 2023 2021]
65%| | 22/34 [00:04<00:02, 5.32it/s]
similar years for 2012 are [2023 2000 2011]
68%| | 23/34 [00:04<00:02, 5.37it/s]
similar years for 2013 are [2016 2020 2024]
71%| | 24/34 [00:04<00:01, 5.36it/s]
similar years for 2014 are [2023 2009 1991]
74%| | 25/34 [00:04<00:01, 5.34it/s]
similar years for 2015 are [2014 2023 1997]
76%| | 26/34 [00:04<00:01, 5.34it/s]
similar years for 2016 are [1998 1995 2007]
79%| | 27/34 [00:05<00:01, 5.32it/s]
similar years for 2017 are [2001 2023 2011]
82%| | 28/34 [00:05<00:01, 5.33it/s]
similar years for 2018 are [2021 2008 2011]
85%| | 29/34 [00:05<00:00, 5.36it/s]
similar years for 2019 are [1998 1995 2023]
88%| | 30/34 [00:05<00:00, 5.40it/s]

similar years for 2020 are [1998 2024 2016]

91%| | 31/34 [00:05<00:00, 5.41it/s]

similar years for 2021 are [2008 1999 2018]

94%| | 32/34 [00:05<00:00, 5.39it/s]

similar years for 2022 are [2018 2010 2011]

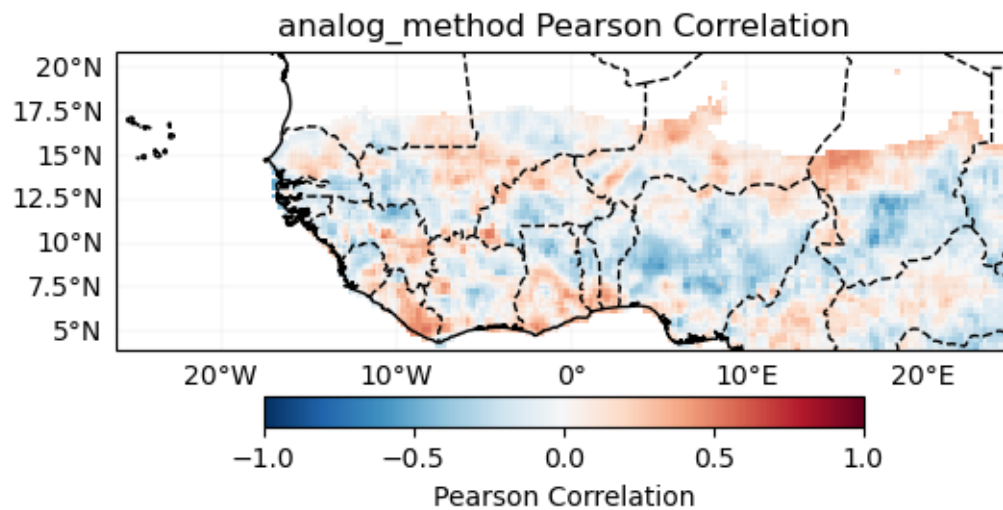
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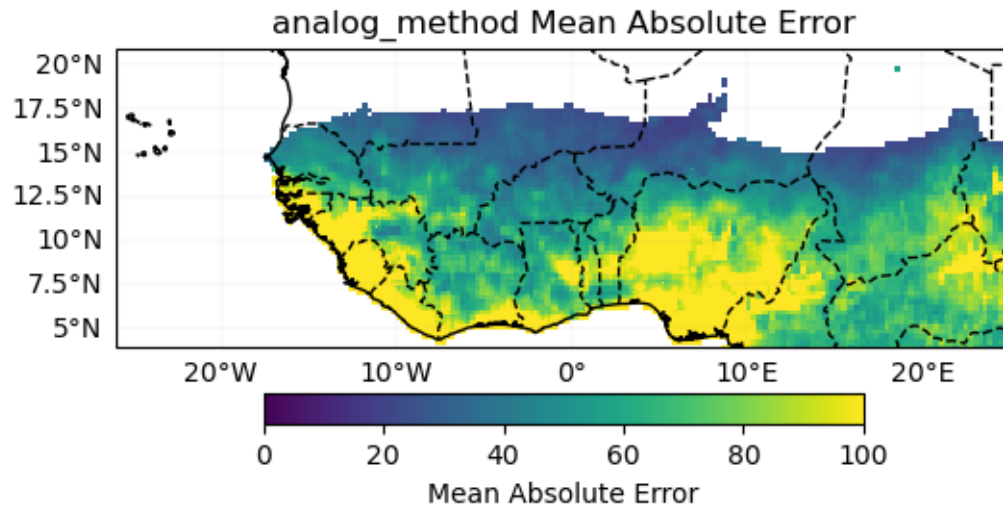
similar years for 2023 are [1997 2012 2011]

100%| | 34/34 [00:06<00:00, 5.35it/s]

similar years for 2024 are [1998 2020 2003]

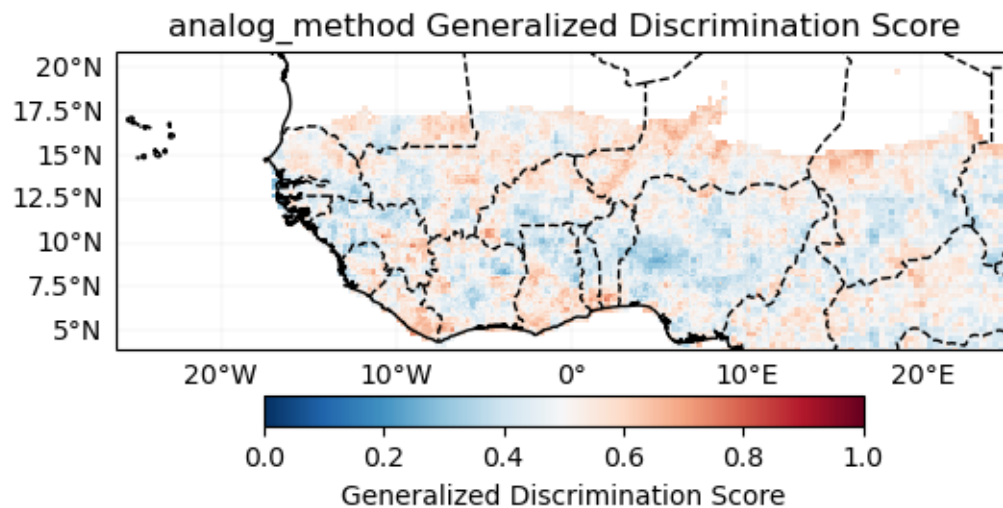
Verification Deterministic





Probabilistic

Generalized Discrimination Score



Forecast

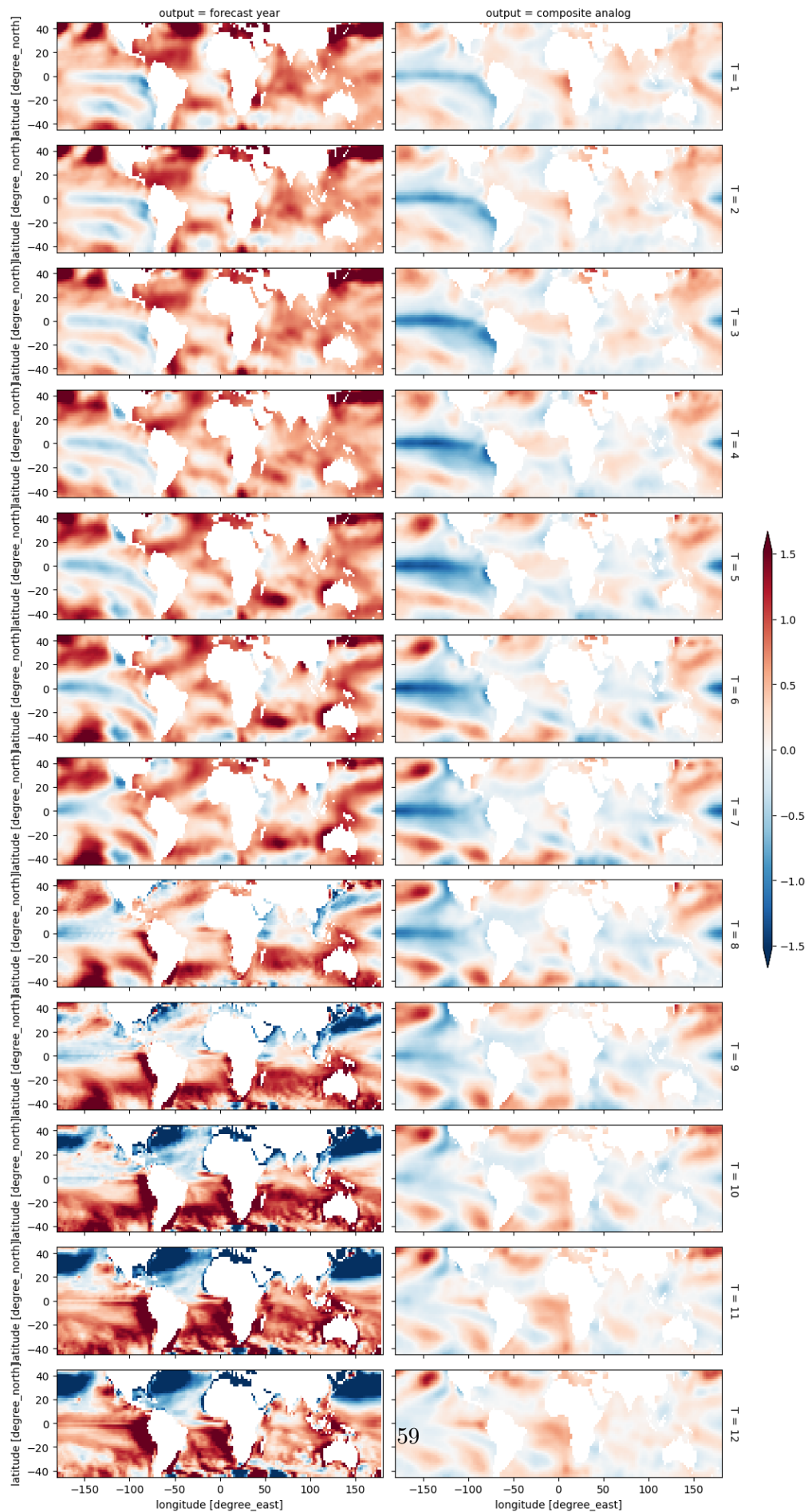
MJJ_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
already exists. Skipping download.

Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
month

MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc

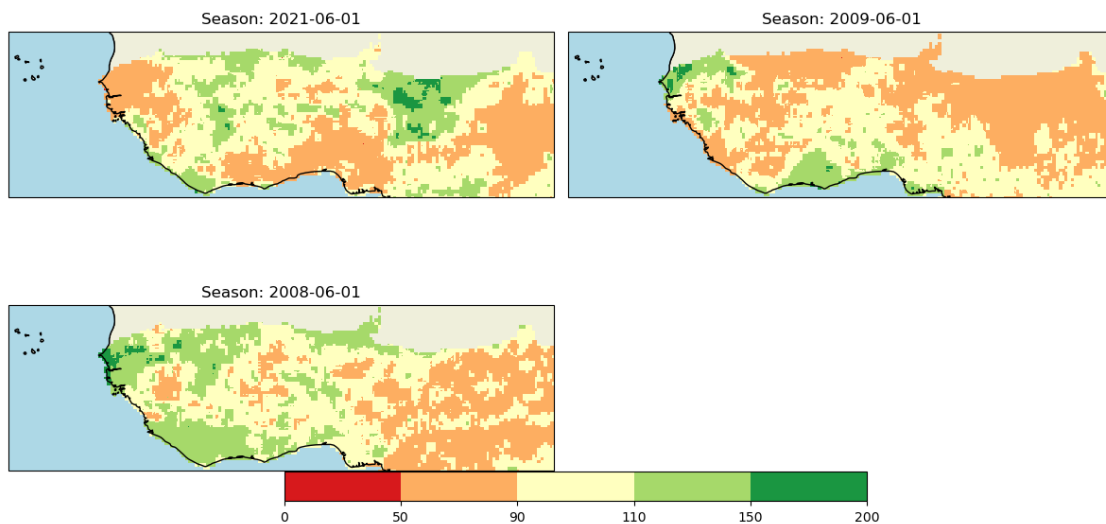
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exists. Skipping download.  
All requested SST data has been processed.  
  
array([2021, 2009, 2008])
```



MJJ_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
 already exists. Skipping download.
 Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
 month
 MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
 File 'MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
 exists. Skipping download.
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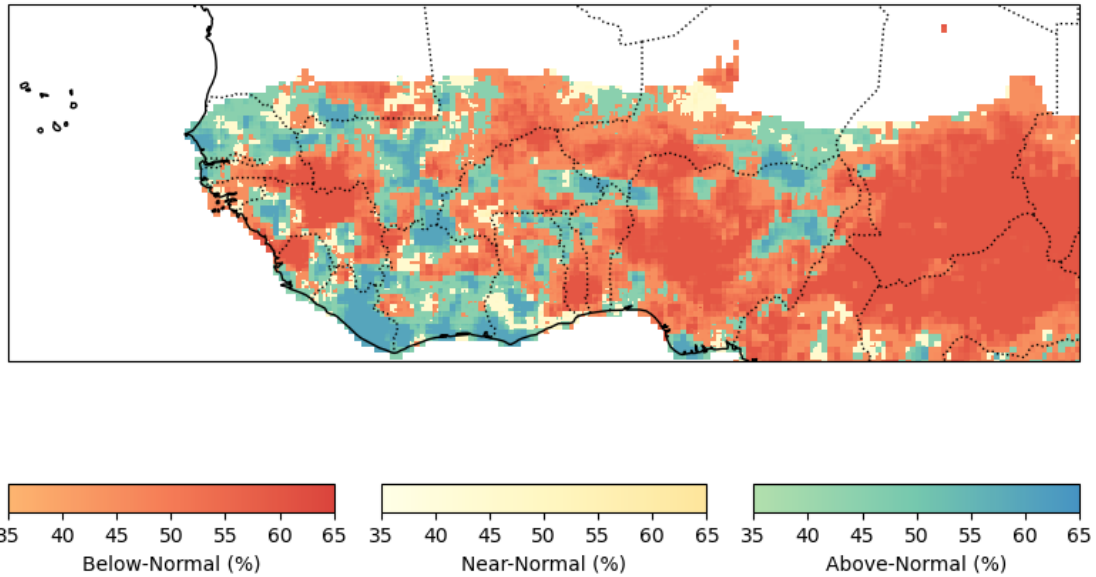
Ratio to Normal Precipitation [%]



array([2021, 2009, 2008])

MJJ_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
 already exists. Skipping download.
 Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
 month
 MJJ_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
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 exists. Skipping download.
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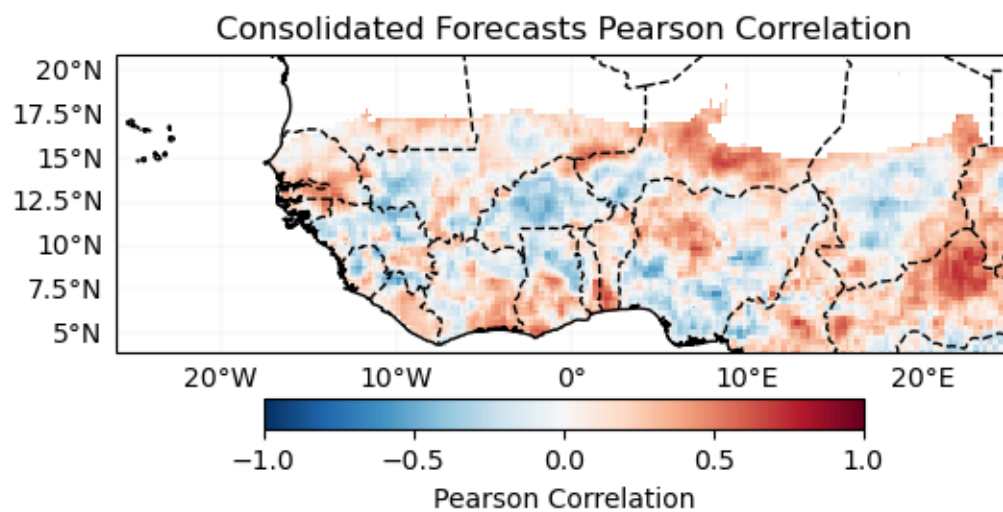
Probabilistic Forecast - analog_method MayJunJul-2025

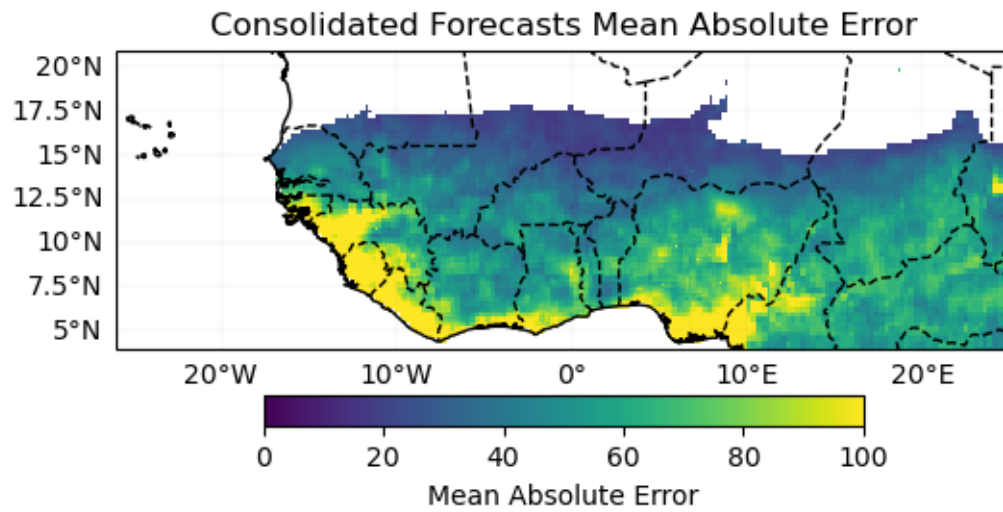


5 Consolidated Forecasts

Cross-validation

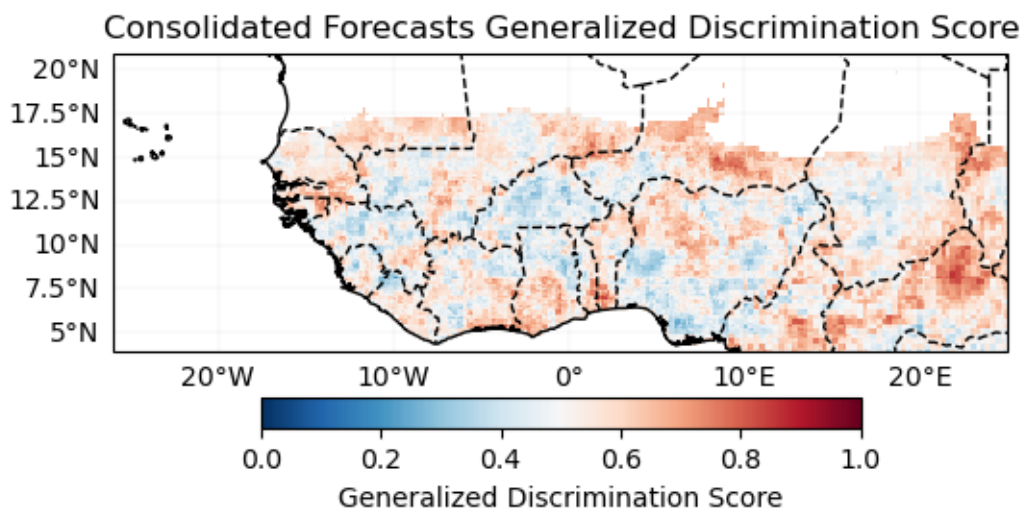
Verification Deterministic





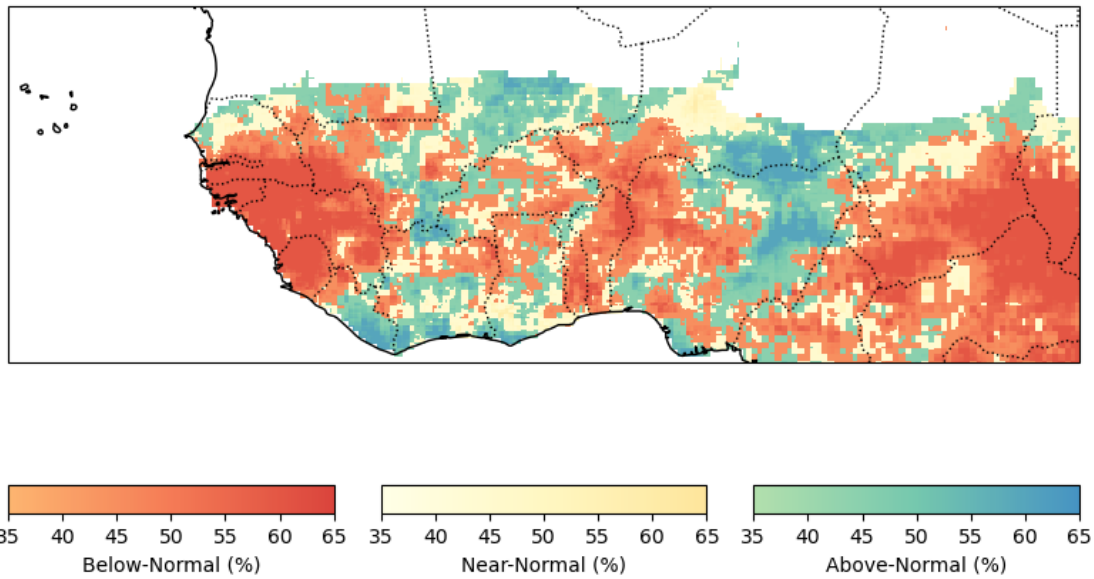
Probabilistic

Generalized Discrimination Score



5.0.1 Forecast

Probabilistic Forecast - Consolidated Forecasts MayJunJul-2025



20713

Save Forecast for future evaluation

5.0.2 END _____

```
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[NbConvertApp] Support files will be in 01_MME_MJJ_2025_files/
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[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 15250394 bytes to 01_MME_MJJ_2025.pdf
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